EC304
Network Theory

Combined Module Notes

Modules 1 – 4

Semester III • ECE • 3 Credits

CGEC | B.Tech ECE (2023–27)

Disclaimer / Study Guide Notice:
These are condensed **Short Notes** focusing on core theory, key formulas, block/circuit diagrams, and standard viva Q&As for university exam revision. Exhaustive numerical practice problems and derivation edge cases are not covered. This serves as a quick-revision reference guide for students.

Biraj Sarkar

IN ASSOCIATION WITH

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Cooch Behar Government Engineering College

Exam-Ready Notes

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Module 1

Circuit Elements and AC Phasor Foundations

1. Circuit Elements and AC Phasor Foundations

Definition: Phasor Transform in Sinusoidal Steady-State

The **phasor transform** maps a time-domain sinusoidal excitation $x(t) = X_m \cos(\omega t + \phi)$ into a frequency-domain complex quantity $\mathbf{X} = X_m e^{j\phi} = X_m \angle \phi$. Under sinusoidal steady-state operating at fixed angular frequency ω, linear differential relations transform into algebraic complex equations:

$$\frac{d}{dt} \longleftrightarrow j\omega, \qquad \int dt \longleftrightarrow \frac{1}{j\omega} \tag{1}$$

1.1 Impedance, Admittance and Reactance of Passive Elements

For passive linear elements, the voltage-current relationship in the frequency domain is governed by Ohm's generalized law: $\mathbf{V} = \mathbf{ZI}$, where $\mathbf{Z} = R + jX$ is the complex **impedance** (Ω), and $\mathbf{Y} = 1/\mathbf{Z} = G + jB$ is the complex **admittance** (S or $\mho$).

Element	Time-Domain Law	Impedance $\mathbf{Z}(j\omega)$	Phase Relationship
Resistor (R)	$v(t) = Ri(t)$	$\mathbf{Z}_R = R\angle 0^\circ$	Voltage and current in exact phase ($\phi = 0$).
Inductor (L)	$v(t) = L\frac{di(t)}{dt}$	$\mathbf{Z}_L = j\omega L = \omega L \angle 90^\circ$	Voltage leads current by 90° ($+\pi/2$ rad).
Capacitor (C)	$i(t) = C\frac{dv(t)}{dt}$	$\mathbf{Z}_C = \frac{1}{j\omega C} = \frac{1}{\omega C}\angle -90^\circ$	Voltage lags current by 90° ($-\pi/2$ rad).

1.2 Controlled (Dependent) Sources

Controlled sources provide a branch voltage or current whose magnitude is linearly proportional to a controlling voltage or current elsewhere in the network:

1. **Voltage-Controlled Voltage Source (VCVS):** $\mathbf{V}_{out} = \mu \mathbf{V}_x$, where μ is dimensionless voltage gain.
2. **Voltage-Controlled Current Source (VCCS):** $\mathbf{I}_{out} = g_m \mathbf{V}_x$, where g_m is mutual transconductance (S).

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3. **Current-Controlled Voltage Source (CCVS):** $\mathbf{V}_{out} = r_m \mathbf{I}_x$, where r_m is mutual transresistance (Ω).
4. **Current-Controlled Current Source (CCCS):** $\mathbf{I}_{out} = \beta \mathbf{I}_x$, where β is dimensionless current gain.

2. Systematic Node and Mesh Analysis (Matrix Approach)

2.1 Matrix Nodal Admittance Formulation

For a network with N independent nodes (excluding the chosen reference/datum node), Kirchhoff's Current Law (KCL) yields a system of N linear simultaneous algebraic equations represented in compact matrix form as:

$$[\mathbf{Y}]\mathbf{V}_n = \mathbf{I}_s \tag{2}$$

where:

$$\begin{bmatrix} Y_{11} & Y_{12} & \cdots & Y_{1N} \\ Y_{21} & Y_{22} & \cdots & Y_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ Y_{N1} & Y_{N2} & \cdots & Y_{NN} \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \\ \vdots \\ \mathbf{V}_N \end{bmatrix} = \begin{bmatrix} \mathbf{I}_{s1} \\ \mathbf{I}_{s2} \\ \vdots \\ \mathbf{I}_{sN} \end{bmatrix} \tag{3}$$

Inspection Rules for Nodal Admittance Matrix [Y]

For networks containing only passive linear elements and independent current sources:

- **Self-Admittance (Y_{ii}):** Sum of all branch admittances directly connected to node i: $Y_{ii} = \sum y_{branch}$.
- **Mutual Admittance ($Y_{ij}, i \neq j$):** Negative of the sum of all branch admittances directly connected between nodes i and j: $Y_{ij} = Y_{ji} = -y_{ij}$.
- **Source Vector ($\mathbf{I}_{si}$):** Net phasor current entering node i from connected independent current sources (entering is positive, leaving is negative).

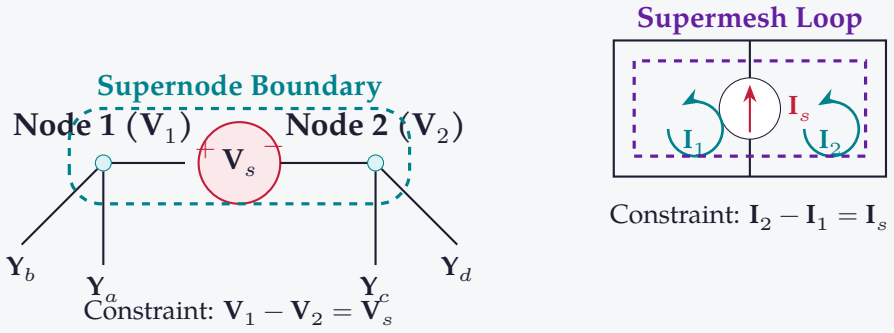
Supernode Technique (Voltage Sources Between Non-Reference Nodes)

When an ideal voltage source (independent or dependent) is connected directly between two non-reference nodes j and k without an intervening series impedance, the current through that source cannot be expressed as $y\Delta V$.

1. **Enclose:** Form a generalized **supernode** by enclosing the voltage source and its two terminating nodes within a single closed Gaussian boundary.
2. **Apply KCL:** Write one KCL equation for the supernode treating all connected external branches.
3. **Constraint Equation:** Write an auxiliary algebraic constraint equation relating the node voltages to the source: $\mathbf{V}_j - \mathbf{V}_k = \mathbf{V}_s$.

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TikZ Illustration: Supernode and Supermesh Topological Concepts



2.2 Matrix Mesh (Loop) Impedance Formulation

For a planar network with M independent meshes, Kirchhoff's Voltage Law (KVL) around clockwise mesh currents yields:

$$[\mathbf{Z}]\mathbf{I}_m = \mathbf{V}_e \tag{4}$$

where:

$$\begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1M} \\ Z_{21} & Z_{22} & \cdots & Z_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ Z_{M1} & Z_{M2} & \cdots & Z_{MM} \end{bmatrix} \begin{bmatrix} \mathbf{I}_1 \\ \mathbf{I}_2 \\ \vdots \\ \mathbf{I}_M \end{bmatrix} = \begin{bmatrix} \mathbf{V}_{e1} \\ \mathbf{V}_{e2} \\ \vdots \\ \mathbf{V}_{eM} \end{bmatrix} \tag{5}$$

Inspection Rules for Mesh Impedance Matrix [Z]

For planar networks with independent voltage sources and passive impedances:

- **Self-Impedance (Z_{ii}):** Total sum of all branch impedances forming mesh i: $Z_{ii} = \sum Z_{loop\,i}$.
- **Mutual Impedance ($Z_{ij}, i \neq j$):** Negative of the branch impedance common to both mesh i and mesh j when mesh currents flow in opposite directions through the shared branch: $Z_{ij} = -Z_{shared}$.
- **Source Vector ($\mathbf{V}_{ei}$):** Algebraic sum of all independent voltage sources encountered in mesh i clockwise (a potential rise yields positive $\mathbf{V}_e$, a potential drop yields negative $\mathbf{V}_e$).

Supermesh Technique (Current Sources Shared Between Meshes)

When an ideal current source is shared between two adjacent meshes i and j:

1. **Avoid Shared Branch:** Create a **supermesh** by conceptually removing the branch containing the current source (leaving an open-circuit path).
2. **Apply KVL:** Write one single KVL equation around the combined outer perimeter loop of both meshes.
3. **Constraint Equation:** Relate the mesh currents to the current source: $\mathbf{I}_j - \mathbf{I}_i = \mathbf{I}_s$.

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3. Source Transformation and Network Duality

3.1 AC Source Transformation

Any practical AC voltage source consisting of an ideal phasor source $\mathbf{V}_s$ in series with internal impedance $\mathbf{Z}_s$ can be transformed into an equivalent practical AC current source consisting of phasor source $\mathbf{I}_s$ in parallel with internal admittance $\mathbf{Y}_s = 1/\mathbf{Z}_s$:

$$\mathbf{I}_s = \frac{\mathbf{V}_s}{\mathbf{Z}_s}, \qquad \mathbf{Z}_s = \frac{1}{\mathbf{Y}_s} \tag{6}$$

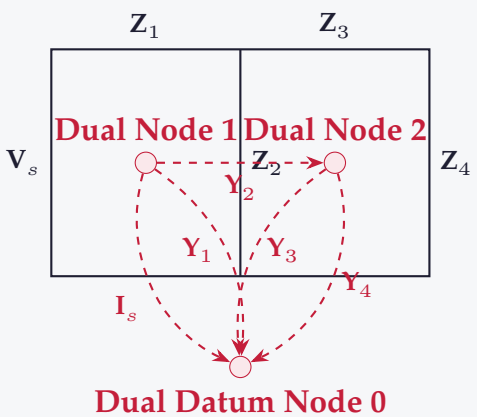
This equivalence holds strictly for all external load terminals at angular frequency ω.

3.2 Network Duality Principles

Two networks N_1 and N_2 are said to be **dual** to each other if the mesh equations of N_1 are mathematically identical in form to the node equations of N_2. Duality is an exact topological and algebraic reflection:

Primary Network Concept	Dual Network Concept	Mathematical Relation
Mesh / Loop	**Node / Node-Pair**	Mesh Current $\mathbf{I}_m \longleftrightarrow$ Node Voltage $\mathbf{V}_n$
Voltage Law (KVL: $\sum \mathbf{V} = 0$**)**	**Current Law (KCL:** $\sum \mathbf{I} = 0$**)**	Dual equations exchange $V \leftrightarrow I$
Series Connection	**Parallel Connection**	$Z = Z_1 + Z_2 \longleftrightarrow Y = Y_1 + Y_2$
Resistance R **(**Ω**)**	**Conductance** G **(S)**	$G = R/R_0^2$ (normalized)
Inductance L **(H)**	**Capacitance** C **(F)**	Numerical value: $C = L/R_0^2$
Voltage Source $\mathbf{V}_s$	**Current Source** $\mathbf{I}_s$	Direction maps clockwise loop $\leftrightarrow$ node entry
Short Circuit $(\mathbf{Z} = 0)$	**Open Circuit** $(\mathbf{Y} = 0)$	$\mathbf{V} = 0 \longleftrightarrow \mathbf{I} = 0$

TikZ Schematic: Dual Network Construction on Planar Graphs



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4. Network Theorems as Applied to AC Circuits

4.1 Superposition Theorem (AC Form)

Superposition Theorem Statement

In any linear, bilateral AC electrical network containing multiple independent sources operating at identical or differing frequencies, the steady-state phasor response (current or voltage) in any branch equals the algebraic sum of the individual phasor responses produced by each independent source acting alone, with all other independent sources set to zero:

- Independent voltage sources are replaced by short circuits ($\mathbf{V}_s = 0$).
- Independent current sources are replaced by open circuits ($\mathbf{I}_s = 0$).
- Dependent sources remain untouched and active throughout the procedure.

Critical Exam Caveat: Superposition with Differing Frequencies and AC Power

1. Multi-Frequency Superposition: If sources operate at distinct angular frequencies ($\omega_1 \neq \omega_2$), phasors **cannot** be added algebraically in the frequency domain because they exist on different complex bases. Responses must be calculated individually as time functions and summed in the time domain:

$$v(t) = v_1(t) + v_2(t) = V_{m1}\cos(\omega_1 t + \phi_1) + V_{m2}\cos(\omega_2 t + \phi_2) \tag{7}$$

2. Average Power is Non-Linear: Superposition does **not** hold for power ($P \neq P_1 + P_2$) when sources share the same frequency, because $P \propto I^2 = (I_1 + I_2)^2 = I_1^2 + I_2^2 + 2I_1 I_2$. Only when sources have orthogonal frequencies ($\omega_1 \neq \omega_2$) does the cross-product average to zero, allowing:

$$P_{\text{avg}} = P_1 + P_2 \qquad (\text{strictly for } \omega_1 \neq \omega_2) \tag{8}$$

4.2 Thevenin's and Norton's Theorems (AC Form)

Thevenin's and Norton's AC Theorems

Thevenin's Theorem: Any linear, bilateral, active two-terminal AC network can be replaced at its external load terminals (A-B) by an equivalent circuit consisting of an open-circuit phasor voltage source $\mathbf{V}_{th}$ connected in series with an internal equivalent impedance $\mathbf{Z}_{th}$.
Norton's Theorem: The same two-terminal AC network can alternatively be replaced by an ideal short-circuit phasor current source $\mathbf{I}_N$ connected in parallel with an internal equivalent admittance $\mathbf{Y}_N = 1/\mathbf{Z}_{th}$.

Analytical Formulation and Evaluation of $\mathbf{V}_{th}$, $\mathbf{I}_N$ and $\mathbf{Z}_{th}$

- **Open-Circuit Voltage ($\mathbf{V}_{th}$):** $\mathbf{V}_{th} = \mathbf{V}_{oc} = \mathbf{V}_A - \mathbf{V}_B$ across open terminals A-B.

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- **Short-Circuit Current ($\mathbf{I}_N$):** $\mathbf{I}_N = \mathbf{I}_{sc}$ flowing from terminal A to B when short-circuited.
- **Source Inter-relation:** $\mathbf{V}_{th} = \mathbf{I}_N \cdot \mathbf{Z}_{th}$.
- **Evaluation of $\mathbf{Z}_{th}$ (Networks with Dependent Sources):** Deactivate all independent sources. Apply an external test source $\mathbf{V}_{\text{test}}$ across terminals A-B and compute resulting input current $\mathbf{I}_{\text{test}}$:

$$\mathbf{Z}_{th} = \frac{\mathbf{V}_{\text{test}}}{\mathbf{I}_{\text{test}}} = \frac{\mathbf{V}_{oc}}{\mathbf{I}_{sc}} \tag{9}$$

4.3 Maximum Power Transfer Theorem in AC Circuits

Let a linear AC source be represented by its Thevenin equivalent ($\mathbf{V}_{th}, \mathbf{Z}_{th} = R_{th} + jX_{th}$) driving a complex load impedance $\mathbf{Z}_L = R_L + jX_L$.

Derivation: Condition for Maximum Average Power Transfer

The phasor current through the load is:

$$\mathbf{I}_L = \frac{\mathbf{V}_{th}}{\mathbf{Z}_{th} + \mathbf{Z}_L} = \frac{\mathbf{V}_{th}}{(R_{th} + R_L) + j(X_{th} + X_L)} \tag{10}$$

The magnitude of the load current squared is:

$$|\mathbf{I}_L|^2 = \frac{|\mathbf{V}_{th}|^2}{(R_{th} + R_L)^2 + (X_{th} + X_L)^2} \tag{11}$$

The real average power dissipated in load resistance R_L is:

$$P_L = |\mathbf{I}_L|^2 R_L = \frac{|\mathbf{V}_{th}|^2 R_L}{(R_{th} + R_L)^2 + (X_{th} + X_L)^2} \tag{12}$$

Step 1: Optimization with respect to X_L: To maximize P_L, the denominator must be minimized. Setting $\frac{\partial P_L}{\partial X_L} = 0$:

$$X_L + X_{th} = 0 \implies \mathbf{X}_L = -\mathbf{X}_{th} \tag{13}$$

The load reactance must cancel the source reactance (conjugate match condition).

Step 2: Optimization with respect to R_L: With $X_L = -X_{th}$, $P_L = \frac{|\mathbf{V}_{th}|^2 R_L}{(R_{th}+R_L)^2}$. Setting $\frac{dP_L}{dR_L} = 0$:

$$\frac{d}{dR_L}\left[\frac{R_L}{(R_{th} + R_L)^2}\right] = \frac{(R_{th} + R_L)^2 - 2R_L(R_{th} + R_L)}{(R_{th} + R_L)^4} = 0 \implies \mathbf{R}_L = \mathbf{R}_{th} \tag{14}$$

Combining both conditions yields the **Complex Conjugate Match**:

$$\mathbf{Z}_L = R_{th} - jX_{th} = \mathbf{Z}_{th}^* \tag{15}$$

The corresponding maximum average power delivered to the load is:

$$P_{\text{max}} = \frac{|\mathbf{V}_{th}|^2 R_{th}}{(2R_{th})^2} = \frac{|\mathbf{V}_{th}|^2}{4R_{th}} \tag{16}$$

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4.3.1 Constrained Load Cases

1. **Load Phase Angle Fixed ($\mathbf{Z}_L = |Z_L|\angle\theta_L$ with variable magnitude):**

$$|Z_L| = |Z_{th}| = \sqrt{R_{th}^2 + X_{th}^2} \tag{17}$$

2. **Purely Resistive Load ($X_L = 0, R_L$ variable):**

$$R_L = |\mathbf{Z}_{th}| = \sqrt{R_{th}^2 + X_{th}^2} \tag{18}$$

3. **Fixed Reactance X_L with Variable Resistance R_L:**

$$R_L = \sqrt{R_{th}^2 + (X_{th} + X_L)^2} \tag{19}$$

4.4 Reciprocity Theorem

Reciprocity Theorem in AC Networks

In any linear, bilateral, passive, time-invariant AC network, if an ideal voltage source $\mathbf{V}_1$ applied in branch 1 produces a current $\mathbf{I}_2$ in branch 2, then the same voltage source transferred to branch 2 will produce identical current $\mathbf{I}_1 = \mathbf{I}_2$ in branch 1:

$$\mathbf{Z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2} = \frac{\mathbf{V}_2}{\mathbf{I}_1} = \mathbf{Z}_{21} \tag{20}$$

The transfer impedance is completely symmetrical ($\mathbf{Z}_{12} = \mathbf{Z}_{21}$).

Restrictions: The network must contain no dependent sources, no unilateral devices (like diodes), and only a single source must be active.

4.5 Compensation (Substitution) Theorem

If the impedance of a branch carrying steady-state phasor current $\mathbf{I}$ changes by an increment $\Delta\mathbf{Z}$ (from $\mathbf{Z}$ to $\mathbf{Z}+\Delta\mathbf{Z}$), the incremental change in currents and voltages throughout all other branches in the network can be calculated by replacing the altered impedance with a virtual compensation voltage source $\mathbf{V}_c$:

$$\mathbf{V}_c = \mathbf{I}'\Delta\mathbf{Z} \approx \mathbf{I}\Delta\mathbf{Z} \tag{21}$$

directed in opposition to the original current $\mathbf{I}$, while deactivating all original independent sources.

4.6 Tellegen's Theorem

Tellegen's Theorem

For any lumped electrical network containing b branches and n nodes, whose branch voltages $v_k(t)$ satisfy KVL and branch currents $i_k(t)$ satisfy KCL under associated

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reference directions:

$$\sum_{k=1}^{b} v_k(t) i_k(t) = 0 \qquad \text{at every instant } t \tag{22}$$

In the frequency domain for AC steady-state sinusoidal circuits:

$$\sum_{k=1}^{b} \mathbf{V}_k \mathbf{I}_k^* = 0 \implies \sum_{k=1}^{b} P_k + j \sum_{k=1}^{b} Q_k = 0 \tag{23}$$

The sum of complex powers delivered to all branches is identically zero (conservation of both active power P and reactive power Q).

Remarkable Scope: Tellegen's theorem is completely independent of the nature of the network components. It holds whether elements are linear, non-linear, passive, active, time-varying, or time-invariant; it relies solely on the topological graph properties governed by KVL and KCL.

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Quick Revision — Module 1

Mesh, Nodal Analysis and AC Network Theorems | EC304

Key Formulas and Network Relations at a Glance

Inductive Reactance	$X_L = \omega L, \quad \mathbf{Z}_L = j\omega L = \omega L \angle 90^\circ$
Capacitive Reactance	$X_C = \frac{1}{\omega C}, \quad \mathbf{Z}_C = \frac{-j}{\omega C} = \frac{1}{\omega C}\angle -90^\circ$
Nodal Admittance Matrix	$[\mathbf{Y}]\mathbf{V}_n = \mathbf{I}_s, \quad Y_{ii} = \sum y_{branch}, \quad Y_{ij} = -y_{ij}$
Mesh Impedance Matrix	$[\mathbf{Z}]\mathbf{I}_m = \mathbf{V}_e, \quad Z_{ii} = \sum Z_{loop}, \quad Z_{ij} = -Z_{shared}$
Source Transformation	$\mathbf{I}_s = \mathbf{V}_s/\mathbf{Z}_s, \quad \mathbf{Z}_s = 1/\mathbf{Y}_s$
Maximum Power Condition (AC)	$\mathbf{Z}_L = \mathbf{Z}_{th}^* = R_{th} - jX_{th}$
Maximum Power Formula	$P_{\max} = \frlac{\lvert\mathbf{V}_{th}\rvert^2}{4R_{th}} \quad (\text{W})$
Resistive Load Max Power	$R_L = \lvert\mathbf{Z}_{th}\rvert = \sqrt{R_{th}^2 + X_{th}^2}$
Reciprocity Transfer Symmetry	$\mathbf{Z}_{12} = \frac{\mathbf{V}_1}{\mathbf{I}_2} = \frac{\mathbf{V}_2}{\mathbf{I}_1} = \mathbf{Z}_{21}$
Tellegen's Complex Conservation	$\sum_{k=1}^{b} \mathbf{V}_k \mathbf{I}_k^* = 0 \iff \sum P_k = 0 \text{ and } \sum Q_k = 0$

One-Line Definitions

- **Phasor:** A complex representation of a sinusoidal voltage or current containing amplitude and phase angle.
- **Impedance (Z):** Total opposition offered by a network element to sinusoidal alternating current, measured in ohms.
- **Admittance (Y):** The measure of ease with which an AC circuit allows current to flow, defined as $1/\mathbf{Z}$.
- **Supernode:** A generalized node created by enclosing a voltage source connected between two non-reference nodes.
- **Supermesh:** A combined loop formed by avoiding a branch containing an ideal current source shared between two meshes.
- **Dual Networks:** Two networks whose loop equations in one are structurally and mathematically identical to the node equations in the other.
- **Thevenin Equivalent:** An AC circuit reduction consisting of an open-circuit voltage $\mathbf{V}_{th}$ in series with $\mathbf{Z}_{th}$.
- **Norton Equivalent:** An AC circuit reduction consisting of a short-circuit current $\mathbf{I}_N$ in parallel with $\mathbf{Z}_{th}$.

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- **Conjugate Match:** Setting load impedance equal to the complex conjugate of source impedance for maximum power transfer.
- **Tellegen's Theorem:** A topological theorem stating that the sum of power delivered over all branches in any lumped network is zero.

Frequently Asked / Viva Questions

Q1. State the condition for maximum power transfer in an AC circuit with complex source and load impedances.

Maximum active power transfer occurs when the load impedance is the complex conjugate of the Thevenin impedance: $\mathbf{Z}_L = \mathbf{Z}_{th}^* = R_{th} - jX_{th}$. Under this condition, reactive power cancels out, and $P_{\max} = \frac{|\mathbf{V}_{th}|^2}{4R_{th}}$.

Q2. Why can the superposition theorem not be directly applied to calculate total AC power?

Because power is a non-linear quadratic function of voltage and current ($P = I^2R$). The square of a sum contains cross-product terms: $(I_1 + I_2)^2 = I_1^2 + I_2^2 + 2I_1I_2 \neq I_1^2 + I_2^2$. Therefore, currents or voltages must be superposed first, and power calculated subsequently from the net total.

Q3. How do you evaluate Thevenin impedance $\mathbf{Z}_{th}$ in a circuit containing dependent sources?

Deactivate all independent sources (short voltage sources, open current sources). Connect a fictitious test source $\mathbf{V}_{\text{test}}$ across the open terminals, determine the resulting current $\mathbf{I}_{\text{test}}$ supplied by the test source, and compute $\mathbf{Z}_{th} = \mathbf{V}_{\text{test}}/\mathbf{I}_{\text{test}}$.

Q4. What are the fundamental requirements for the Reciprocity theorem to hold in a network?

The network must be linear, passive, time-invariant, bilateral, and contain no dependent sources or non-linear elements. Only a single excitation source must be active at a time.

Q5. What is the physical meaning of Tellegen's theorem in an AC circuit?

It proves that in any lumped network, the sum of all instantaneous branch powers equals zero. In AC circuits, this implies the simultaneous conservation of both active real power ($\sum P_k = 0$) and reactive imaginary power ($\sum Q_k = 0$).

Q6. When is a supernode required in nodal analysis?

When an ideal voltage source (independent or dependent) is connected directly between two non-reference nodes without any series impedance, making the branch current indeterminate from node voltages alone.

Q7. Explain the concept of duality in electrical networks.

Two planar networks are duals if the KVL loop equations of one match the KCL node equations of the other. Series branches map to parallel branches, meshes map to nodes, inductors map to capacitors, and voltage sources map to current sources.

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Frequently Asked / Viva Questions (contd.)

Q8. If the load is purely resistive ($X_L = 0$), what is the condition for maximum power transfer from an AC source?

The load resistance must equal the magnitude of the source Thevenin impedance: $R_L = |\mathbf{Z}_{th}| = \sqrt{R_{th}^2 + X_{th}^2}$.

Q9. Can Tellegen's theorem be applied to non-linear and time-varying circuits?

Yes. Tellegen's theorem depends strictly on Kirchhoff's laws (KVL and KCL), which stem solely from topology (conservation of energy and charge). It holds irrespective of whether elements are linear, non-linear, active, passive, or time-varying.

Q10. What is the difference between Mesh Analysis and Loop Analysis?

A mesh is a loop that does not contain any other loops within it (applicable strictly to planar circuits). Loop analysis can be applied to arbitrary closed paths and non-planar networks using fundamental loop sets.

Summary Matrix: Comprehensive Comparison of AC Network Theorems

Theorem	Primary Condition	Practical Applications and Limitations
Superposition	**Linear, bilateral elements**	Multi-frequency circuits; invalid for power calculations.
Thevenin's	**Linear active two-terminal**	Simplifies terminal analysis; $\mathbf{Z}_{th}$ needs test source if dependent sources exist.
Norton's	**Dual of Thevenin's theorem**	Current-driven systems; relates via $\mathbf{V}_{th} = \mathbf{I}_N \mathbf{Z}_{th}$.
Maximum Power Transfer	**Source $\mathbf{Z}_{th}$ fixed, load adjustable**	Conjugate match $\mathbf{Z}_L = \mathbf{Z}_{th}^*$; efficiency is strictly 50%.
Reciprocity	**Linear, bilateral, passive, no dependent sources**	Symmetrical transfer impedance; useful in antennas and filters.
Tellegen's	**Any lumped network obeying KVL & KCL**	Holds for linear/non-linear, active/passive; confirms energy conservation.

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Module 2

Fourier Series Analysis in Electrical Networks

1. Fourier Series Analysis in Electrical Networks

Definition: Trigonometric and Exponential Fourier Series

Any physically realizable periodic electrical signal $f(t) = f(t+T)$ with fundamental period T and fundamental angular frequency $\omega_0 = 2\pi/T$ satisfying the Dirichlet conditions can be decomposed into an infinite sum of harmonically related sinusoids:

1. **Trigonometric Fourier Series (TFS):**

$$f(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \tag{24}$$

where the Fourier coefficients are defined over any arbitrary period t_0 to $t_0 + T$:

$$a_0 = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)\,dt, \qquad a_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\cos(n\omega_0 t)\,dt \tag{25}$$

$$b_n = \frac{2}{T}\int_{t_0}^{t_0+T} f(t)\sin(n\omega_0 t)\,dt \tag{26}$$

2. **Compact (Harmonic) Trigonometric Form:**

$$f(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t - \theta_n), \quad c_0 = a_0, \quad c_n = \sqrt{a_n^2 + b_n^2}, \quad \theta_n = \tan^{-1}\left(\frac{b_n}{a_n}\right) \tag{27}$$

3. **Exponential Fourier Series (EFS):**

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \qquad C_n = \frac{1}{T}\int_{t_0}^{t_0+T} f(t)e^{-jn\omega_0 t}dt \tag{28}$$

1.1 Inter-relationship Between Coefficients

The complex exponential and real trigonometric coefficients relate directly via Euler's identities:

$$C_0 = a_0, \qquad C_n = \frac{a_n - jb_n}{2} = \frac{c_n}{2}e^{-j\theta_n}, \qquad C_{-n} = C_n^* = \frac{a_n + jb_n}{2} \tag{29}$$

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Conversely:

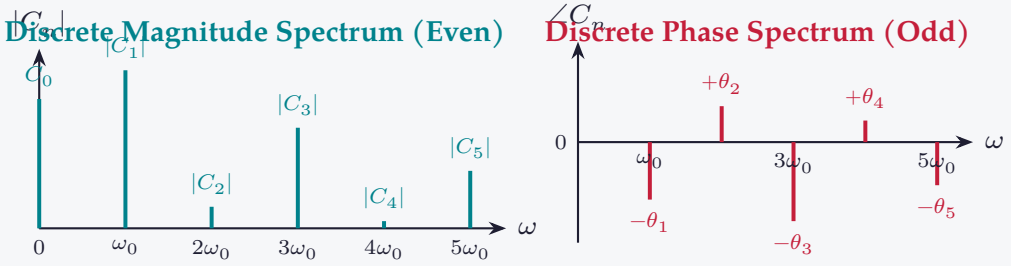
$$a_n = C_n + C_{-n} = 2\text{Re}\{C_n\}, \qquad b_n = j(C_n - C_{-n}) = -2\text{Im}\{C_n\} \tag{30}$$

1.2 Waveform Symmetry and Harmonic Content

Exploiting waveform symmetry simplifies Fourier integration and provides immediate diagnostic criteria for harmonic absence:

Symmetry Type	Condition	Zero Coefficients	Surviving Harmonics
Even Symmetry	$f(-t) = f(t)$	$b_n = 0 \quad \forall n$	DC component and cosine harmonics only.
Odd Symmetry	$f(-t) = -f(t)$	$a_0 = 0, \quad a_n = 0 \quad \forall n$	Sine harmonics only (strictly no DC).
Half-Wave (HWS)	$f(t \pm \frac{T}{2}) = -f(t)$	$a_0 = 0,\ a_{2k} = b_{2k} = 0$	Strictly odd harmonics only $(n = 1, 3, 5, \dots)$.
Even Quarter-Wave	**HWS + Even**	$a_0 = 0,\ b_n = 0 \quad \forall n$	Odd cosine harmonics only (a_{2k+1}).
Odd Quarter-Wave	**HWS + Odd**	$a_0 = 0,\ a_n = 0 \quad \forall n$	Odd sine harmonics only (b_{2k+1}).

TikZ Illustration: Discrete Line Spectra of a Periodic Signal



2. Network Response to Non-Sinusoidal Periodic Inputs

2.1 Harmonic Superposition Method

When a linear time-invariant circuit is excited by a non-sinusoidal periodic voltage source:

$$v(t) = V_0 + \sum_{n=1}^{\infty} V_{mn} \cos(n\omega_0 t + \theta_{vn}) \tag{31}$$

the steady-state current response is calculated by applying the **Superposition Principle** to each harmonic component independently.

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Systematic Harmonic Solution Procedure

1. **DC Analysis ($n = 0$):** Replace inductors with short circuits ($\mathbf{Z}_L = 0$) and capacitors with open circuits ($\mathbf{Z}_C = \infty$). Compute DC current:

$$I_0 = \frac{V_0}{R_{\text{eq}}} \tag{32}$$

2. **n-th Harmonic Impedance:** Evaluate network input impedance at harmonic frequency $\omega_n = n\omega_0$:

$$\mathbf{Z}(jn\omega_0) = R_n + jX_n = |\mathbf{Z}_n|e^{j\phi_n} = |\mathbf{Z}_n|\angle\phi_n \tag{33}$$

where inductive reactance scales directly ($X_{Ln} = n\omega_0 L$) and capacitive reactance scales inversely ($X_{Cn} = \frac{1}{n\omega_0 C}$).

3. **n-th Harmonic Current Phasor:**

$$\mathbf{I}_{mn} = \frac{\mathbf{V}_{mn}\angle\theta_{vn}}{|\mathbf{Z}_n|\angle\phi_n} = \frac{V_{mn}}{|\mathbf{Z}_n|}\angle(\theta_{vn} - \phi_n) = I_{mn}\angle\theta_{in} \tag{34}$$

4. **Total Time-Domain Current Synthesis:**

$$i(t) = I_0 + \sum_{n=1}^{\infty} I_{mn}\cos(n\omega_0 t + \theta_{vn} - \phi_n) \tag{35}$$

2.2 Effective (RMS) Values and Average Power

Formulas for RMS Values and Non-Sinusoidal Power

- **True RMS (Effective) Voltage and Current:**

$$V_{\text{rms}} = \sqrt{\frac{1}{T}\int_0^T v^2(t)dt} = \sqrt{V_0^2 + \sum_{n=1}^{\infty}\frac{V_{mn}^2}{2}} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} V_{n,\text{rms}}^2} \tag{36}$$

$$I_{\text{rms}} = \sqrt{I_0^2 + \sum_{n=1}^{\infty} I_{n,\text{rms}}^2} \tag{37}$$

- **Total Average Active Power (P):** By orthogonality of distinct sinusoids over fundamental period T, all cross-harmonic terms integrate to zero:

$$P = \frac{1}{T}\int_0^T v(t)i(t)dt = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}}\cos(\phi_n) \tag{38}$$

- **Apparent Power (S), Reactive Power (Q) and Distortion Power (D):**

$$S = V_{\text{rms}} I_{\text{rms}}, \qquad Q = \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}}\sin(\phi_n), \qquad D = \sqrt{S^2 - P^2 - Q^2} \tag{39}$$

- **True Power Factor and Distortion Factor:**

$$\text{PF} = \frac{P}{S} = \frac{I_{1,\text{rms}}}{I_{\text{rms}}} \cdot \cos\phi_1 = (\text{Distortion Factor}) \times (\text{Displacement Factor}) \tag{40}$$

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- **Total Harmonic Distortion (THD):**

$$\text{THD}_I = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}} = \sqrt{\left(\frac{I_{\text{rms}}}{I_{1,\text{rms}}}\right)^2 - 1} \tag{41}$$

3. The Continuous Spectrum via Fourier Transform

Continuous Fourier Transform Pair

As period $T \to \infty$, discrete spectral lines merge into a continuous density spectrum:

$$X(j\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt \tag{42}$$

$$x(t) = \mathcal{F}^{-1}\{X(j\omega)\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega \tag{43}$$

The **Energy Spectral Density** is $\mathcal{E}(\omega) = |X(j\omega)|^2$. Total signal energy is conserved under the transformation via **Parseval's Energy Theorem**:

$$W = \int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega \tag{44}$$

4. Three-Phase Unbalanced Circuits and Symmetrical Components

4.1 Symmetrical Components Transformation (Fortescue's Theorem)

In an unbalanced three-phase system, the three unsymmetrical phasor voltages or currents can be decomposed into three balanced symmetrical components:

Fortescue's Theorem and the Complex Operator a

Any set of three unbalanced phasors $\mathbf{V}_a, \mathbf{V}_b, \mathbf{V}_c$ can be uniquely expressed as the sum of three balanced symmetrical sets:

1. **Positive-Sequence ($\mathbf{V}_{a1}, \mathbf{V}_{b1}, \mathbf{V}_{c1}$):** Equal magnitudes, 120° phase separation, same phase sequence (a-b-c) as original.
2. **Negative-Sequence ($\mathbf{V}_{a2}, \mathbf{V}_{b2}, \mathbf{V}_{c2}$):** Equal magnitudes, 120° phase separation, reversed phase sequence (a-c-b).
3. **Zero-Sequence ($\mathbf{V}_{a0}, \mathbf{V}_{b0}, \mathbf{V}_{c0}$):** Equal magnitudes and identical phase angles (in-phase).

Defining the complex phase-shift operator $a = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}$:

$$a^2 = e^{j240^\circ} = -\frac{1}{2} - j\frac{\sqrt{3}}{2}, \qquad a^3 = 1, \qquad 1 + a + a^2 = 0 \tag{45}$$

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Matrix Transformation Relations

The sequence components relate to the phase quantities by:

$$\begin{bmatrix} \mathbf{V}_a \\ \mathbf{V}_b \\ \mathbf{V}_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} \mathbf{V}_{a0} \\ \mathbf{V}_{a1} \\ \mathbf{V}_{a2} \end{bmatrix} = [\mathbf{A}]\mathbf{V}_{012} \tag{46}$$

The inverse transformation matrix $[\mathbf{A}]^{-1}$ yields:

$$\begin{bmatrix} \mathbf{V}_{a0} \\ \mathbf{V}_{a1} \\ \mathbf{V}_{a2} \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \mathbf{V}_a \\ \mathbf{V}_b \\ \mathbf{V}_c \end{bmatrix} \tag{47}$$

Explicitly:

$$\mathbf{V}_{a0} = \frac{\mathbf{V}_a + \mathbf{V}_b + \mathbf{V}_c}{3}, \quad \mathbf{V}_{a1} = \frac{\mathbf{V}_a + a\mathbf{V}_b + a^2\mathbf{V}_c}{3}, \quad \mathbf{V}_{a2} = \frac{\mathbf{V}_a + a^2\mathbf{V}_b + a\mathbf{V}_c}{3} \tag{48}$$

Physical Consequence: Zero-Sequence Current in Three-Phase Circuits

In a three-phase system, the neutral current $\mathbf{I}_n$ is the sum of line currents:

$$\mathbf{I}_n = \mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 3\mathbf{I}_{a0} \tag{49}$$

Conclusion:

- In a 3-phase 3-wire system (ungrounded), $\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0 \implies \mathbf{I}_{a0} = 0$. Zero-sequence line current **cannot** flow in any 3-wire system.
- In delta-connected loads, line voltages sum to zero ($\mathbf{V}_{ab} + \mathbf{V}_{bc} + \mathbf{V}_{ca} = 0 \implies \mathbf{V}_{L0} = 0$), so zero-sequence voltage cannot exist across lines.

4.2 Neutral Shift Voltage (Millman's Theorem in 3-Phase Systems)

When an unbalanced star-connected load with phase admittances $\mathbf{Y}_a, \mathbf{Y}_b, \mathbf{Y}_c$ is connected to a balanced source with neutral N, the star load neutral point N' floats to a potential known as the **neutral shift voltage** $\mathbf{V}_{N'N}$:

$$\mathbf{V}_{N'N} = \frac{\mathbf{V}_{AN}\mathbf{Y}_a + \mathbf{V}_{BN}\mathbf{Y}_b + \mathbf{V}_{CN}\mathbf{Y}_c}{\mathbf{Y}_a + \mathbf{Y}_b + \mathbf{Y}_c + \mathbf{Y}_n} \tag{50}$$

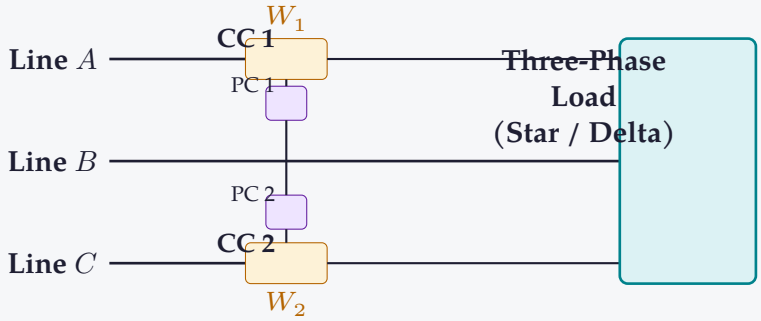
where $\mathbf{Y}_n$ is the neutral wire admittance ($\mathbf{Y}_n = 0$ for 3-wire floating neutral). The voltage across each load branch is then $\mathbf{V}_{aN'} = \mathbf{V}_{AN} - \mathbf{V}_{N'N}$.

4.3 Power Measurement by Two-Wattmeter Method

The two-wattmeter method measures total active power in any 3-phase 3-wire system (balanced or unbalanced, star or delta):

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TikZ Schematic: Two-Wattmeter Connection in a Three-Phase System



Two-Wattmeter Power and Power Factor Relations

Under balanced sinusoidal conditions with line-to-line voltage V_L, line current I_L, and load power factor angle ϕ:

$$W_1 = V_L I_L \cos(30^\circ - \phi), \qquad W_2 = V_L I_L \cos(30^\circ + \phi) \tag{51}$$

1. **Total Active Power (P):**

$$P = W_1 + W_2 = V_L I_L[\cos(30^\circ - \phi) + \cos(30^\circ + \phi)] = \sqrt{3} V_L I_L \cos\phi \tag{52}$$

2. **Total Reactive Power (Q):**

$$W_1 - W_2 = V_L I_L[\cos(30^\circ - \phi) - \cos(30^\circ + \phi)] = V_L I_L \sin\phi \implies Q = \sqrt{3}(W_1 - W_2) \tag{53}$$

3. **Power Factor Angle (ϕ):**

$$\tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2} \implies \text{PF} = \cos\phi = \cos\left[\tan^{-1}\left(\sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}\right)\right] \tag{54}$$

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Quick Revision — Module 2

Fourier Analysis and Three-Phase Unbalanced Systems | EC304

Key Formulas and Network Relations at a Glance

Trigonometric Fourier Form	$f(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$
Exponential Fourier Form	$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}, \quad C_n = \frac{a_n - jb_n}{2}$
True RMS Voltage	$V_{\text{rms}} = \sqrt{V_0^2 + \sum_{n=1}^{\infty} V_{n,\text{rms}}^2}$
Average Non-Sinusoidal Power	$P = V_0 I_0 + \sum_{n=1}^{\infty} V_{n,\text{rms}} I_{n,\text{rms}} \cos(\phi_n)$
True Power Factor	$\text{PF} = \frac{P}{S} = \frac{I_{1,\text{rms}}}{I_{\text{rms}}} \cos\phi_1$
Total Harmonic Distortion	$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} I_{n,\text{rms}}^2}}{I_{1,\text{rms}}}$
Complex Operator a	$a = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2}, \quad 1 + a + a^2 = 0$
Zero-Sequence Current	$\mathbf{I}_{a0} = \frac{\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c}{3} = \frac{\mathbf{I}_n}{3}$
Two-Wattmeter Total Power	$P_{\text{total}} = W_1 + W_2 = \sqrt{3} V_L I_L \cos\phi$
Two-Wattmeter Power Factor	$\tan\phi = \sqrt{3}\frac{W_1 - W_2}{W_1 + W_2}$

One-Line Definitions

- **Harmonic Frequency:** An integer multiple ($n\omega_0$) of the fundamental repetition frequency of a periodic wave.
- **Half-Wave Symmetry:** Condition $f(t \pm T/2) = -f(t)$, ensuring complete cancellation of all even harmonic orders.
- **Line Spectrum:** A discrete graphical plot of harmonic component amplitudes and phase shifts versus frequency.
- **Harmonic Impedance:** The frequency-dependent impedance of a circuit evaluated specifically at harmonic frequency $n\omega_0$.
- **Distortion Power (D):** The component of apparent power caused strictly by harmonic cross-frequencies, measured in VA.
- **Displacement Factor:** The cosine of the fundamental phase angle ($\cos\phi_1$), measuring fundamental displacement.
- **Symmetrical Components:** Three balanced phasor sets (positive, negative, zero) replacing an unbalanced three-phase system.
- **Neutral Shift Voltage:** The potential difference appearing between floating load neutral and source star neutral in unbalanced loads.
- **Two-Wattmeter Method:** A standard method using two power meters to measure total power in any three-phase three-wire system.

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- **Parseval's Theorem:** Mathematical relation stating total energy is conserved between time and continuous frequency domains.

Frequently Asked / Viva Questions

Q1. What harmonics are present in a periodic waveform possessing Half-Wave Symmetry?

A waveform with half-wave symmetry ($f(t \pm T/2) = -f(t)$) contains strictly odd harmonics ($n = 1, 3, 5, \dots$). The DC component (a_0) and all even harmonics (a_{2k}, b_{2k}) are identically zero.

Q2. Why does a non-sinusoidal voltage produce an active power contribution only with currents of the identical harmonic order?

Because the time-average product of two sinusoids of differing frequencies over the fundamental period is identically zero: $\int_0^T \cos(m\omega_0 t)\cos(n\omega_0 t)dt = 0$ for $m \neq n$. Active power is generated exclusively by voltage and current terms belonging to the same harmonic frequency.

Q3. Can zero-sequence currents flow in a balanced or unbalanced 3-phase 3-wire system? Explain.

No. In any 3-phase 3-wire system, by KCL, the sum of line currents must be zero ($\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c = 0$). Since zero-sequence current is $\mathbf{I}_{a0} = \frac{1}{3}(\mathbf{I}_a + \mathbf{I}_b + \mathbf{I}_c)$, it is strictly zero. A neutral or ground path is required for zero-sequence current to exist.

Q4. What does it mean if one wattmeter reads negative in the two-wattmeter method?

A negative reading occurs when the load power factor is less than 0.5 ($\phi > 60^\circ$). In that case, $W_2 = V_L I_L \cos(30^\circ + \phi)$ becomes negative. Total power is still the algebraic sum $W_1 + W_2$.

Q5. What are the readings of the two wattmeters at unity power factor ($\cos\phi = 1$)?

At unity power factor, $\phi = 0^\circ$. Thus, $W_1 = V_L I_L \cos(30^\circ) = \frac{\sqrt{3}}{2} V_L I_L$ and $W_2 = V_L I_L \cos(30^\circ) = \frac{\sqrt{3}}{2} V_L I_L$. Both wattmeters read identical positive values ($W_1 = W_2 = P/2$).

Q6. What are the readings of the two wattmeters at zero power factor ($\cos\phi = 0$)?

At zero power factor, $\phi = 90^\circ$. Then $W_1 = V_L I_L \cos(-60^\circ) = 0.5 V_L I_L$ and $W_2 = V_L I_L \cos(120^\circ) = -0.5 V_L I_L$. The wattmeters have equal magnitudes but opposite signs ($W_1 = -W_2$), yielding net active power $P = W_1 + W_2 = 0$.

Q7. State the relationship between Total Harmonic Distortion (THD) and Distortion Factor.

The distortion factor $\mu = I_{1,\text{rms}}/I_{\text{rms}}$ relates to THD by $\mu = \frac{1}{\sqrt{1+\text{THD}^2}}$. As harmonic distortion increases, the distortion factor decreases below unity, degrading true power factor.

Q8. What is the physical meaning of the complex operator $a = e^{j120^\circ}$?

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Frequently Asked / Viva Questions (contd.)

The operator a represents a counter-clockwise phase rotation of 120° ($2\pi/3$ radians) without altering the magnitude of the phasor. Multiplying by a^2 rotates by 240°.

Q9. How does Millman's Theorem find the neutral shift in an unbalanced 3-phase star circuit?

Millman's theorem treats the 3 phase voltages as parallel branches connected to neutral. The star point voltage is $\mathbf{V}_{N'N} = (\sum \mathbf{V}_k \mathbf{Y}_k)/\sum \mathbf{Y}_k$, directly yielding the shift in neutral potential.

Q10. Distinguish between continuous and discrete Fourier spectra.

A discrete spectrum consists of isolated impulses at integer multiples of fundamental frequency $n\omega_0$, characteristic of periodic signals. A continuous spectrum is a continuous function of ω spanning all frequencies, characteristic of non-periodic transient energy pulses.

Summary Matrix: Waveform Symmetries and Two-Wattmeter Diagnostic Rules

Operating Condition	Wattmeter Relationship	Power Factor / Significance
$\phi = 0^\circ$ **(Resistive)**	$W_1 = W_2 > 0$	PF $= 1.0$; identical positive readings.
$\phi = 60^\circ$	$W_1 > 0, \quad W_2 = 0$	PF $= 0.5$; meter 2 reads exactly zero.
$\phi > 60^\circ$ **(Lagging)**	$W_1 > 0, \quad W_2 < 0$	PF < 0.5; meter 2 reverses deflection.
$\phi = 90^\circ$ **(Pure Reactive)**	$W_1 = -W_2$	PF $= 0$; net active power is strictly zero.
Unbalanced 3-Wire System	$W_1 + W_2 = P_{\text{total}}$	Valid regardless of load unbalance or power factor.

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Module 3

Singularity Functions and Waveform Synthesis

1. Singularity Functions and Waveform Synthesis

Singularity functions (also referred to as switching functions or test signals) are a family of mathematical idealizations exhibiting discontinuities or discontinuous derivatives. In linear circuit theory, they model switching actions, sharp current/voltage pulses, and sudden transients.

1.1 Standard Singularity Functions

Definitions of Core Singularity Functions

1. **Unit Step Function** $u(t)$**:** Models the instantaneous closing or opening of an ideal switch at $t = 0$:

$$u(t) = \begin{cases} 0, & t < 0 \\ 1, & t > 0 \end{cases} \qquad \text{Delayed: } u(t - t_0) = \begin{cases} 0, & t < t_0 \\ 1, & t > t_0 \end{cases} \tag{55}$$

2. **Unit Impulse (Dirac Delta) Function** $\delta(t)$**:** Defined as the distributional derivative of the unit step function:

$$\delta(t) = \frac{du(t)}{dt}, \quad \delta(t) = 0 \text{ for } t \neq 0, \quad \int_{-\infty}^{\infty} \delta(t)\, dt = 1 \tag{56}$$

Sifting (Sampling) Property:

$$\int_{-\infty}^{\infty} f(t)\, \delta(t - t_0)\, dt = f(t_0) \tag{57}$$

3. **Unit Ramp Function** $r(t)$**:** Represents a linearly growing excitation starting at $t = 0$:

$$r(t) = \int_{-\infty}^{t} u(\tau)\, d\tau = t\, u(t) = \begin{cases} 0, & t < 0 \\ t, & t \geq 0 \end{cases} \tag{58}$$

4. **Unit Doublet** $\delta'(t)$**:** The time derivative of the unit impulse function:

$$\delta'(t) = \frac{d\delta(t)}{dt}, \qquad \int_{-\infty}^{\infty} f(t)\, \delta'(t - t_0)\, dt = -f'(t_0) \tag{59}$$

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Hierarchy of Singularity Functions

The standard singularity functions form a continuous calculus chain linked by differentiation and integration:

$$\cdots \xleftarrow{\frac{d}{dt}} \delta'(t) \xleftarrow{\frac{d}{dt}} \delta(t) \xleftarrow{\frac{d}{dt}} u(t) \xleftarrow{\frac{d}{dt}} r(t) \xleftarrow{\frac{d}{dt}} \frac{t^2}{2}u(t) \cdots \tag{60}$$

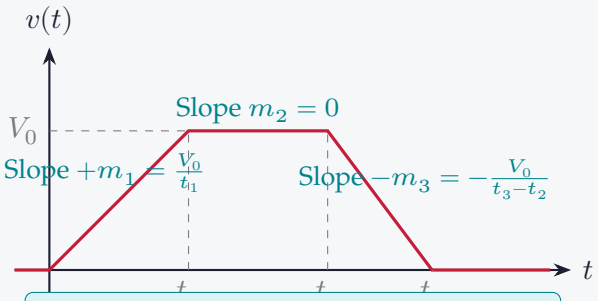
1.2 Waveform Synthesis Technique

Any arbitrary piecewise-linear or stepped signal can be synthesized systematically as a linear algebraic sum of delayed ramp functions and delayed step functions:

$$v(t) = \sum_{k=0}^{N} \Delta m_k \, r(t - t_k) + \sum_{j=0}^{M} \Delta v_j \, u(t - t_j) \tag{61}$$

where $\Delta m_k = m_k - m_{k-1}$ is the change in slope at breakpoint t_k, and Δv_j is the abrupt jump in amplitude at breakpoint t_j.

TikZ Schematic: Synthesis of a Trapezoidal Pulse from Singularity Functions



Synthesis Equation:
$v(t) = m_1 \, r(t) - m_1 \, r(t - t_1) - m_3 \, r(t - t_2) + m_3 \, r(t - t_3)$

Step-by-Step Example: Waveform Synthesis and its Transform

Problem: Express a triangular pulse that rises linearly from 0 to 10 V between $t = 0$ and $t = 2$ s, and drops instantaneously to zero at $t = 2$ s. Determine its Laplace transform.
Solution:

1. Between $t = 0$ and $t = 2$, the waveform has a constant slope $m_1 = \frac{10-0}{2-0} = +5$ V/s. It starts at $t = 0$, represented as $5\,r(t)$.
2. At $t = 2$ s, the ramp must be cancelled by subtracting an identical ramp of slope 5: $-5\,r(t-2)$.
3. At $t = 2$ s, the voltage must also jump downwards instantaneously by -10 V to reach 0: $-10\,u(t-2)$.
4. Synthesized time-domain expression:

$$v(t) = 5\,r(t) - 5\,r(t-2) - 10\,u(t-2) \tag{62}$$

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5. Applying the time-shifting property of the Laplace transform:

$$V(s) = \frac{5}{s^2} - \frac{5}{s^2}\,e^{-2s} - \frac{10}{s}\,e^{-2s} = \frac{5}{s^2}\left[1 - e^{-2s}(1+2s)\right] \tag{63}$$

2. The Laplace Transform and Operational Theorems

The unilateral (one-sided) Laplace transform is an essential mathematical tool for solving linear, time-invariant (LTI) differential equations with non-zero initial boundary conditions.

2.1 Definition and Region of Convergence

Unilateral Laplace Transform

Let $f(t)$ be a continuous or piecewise-continuous function defined for $t \geq 0$. The unilateral Laplace transform $\mathcal{L}\{f(t)\} = F(s)$ is:

$$F(s) = \int_{0^-}^{\infty} f(t)\,e^{-st}\,dt \tag{64}$$

where $s = \sigma + j\omega$ is the complex frequency variable. The lower integration limit $0^- \equiv \lim_{\epsilon \to 0^+}(0-\epsilon)$ ensures that singularity functions (such as $\delta(t)$) acting at $t = 0$ are rigorously integrated and accounted for.

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2.2 Operational Theorems for Circuit Analysis

Property / Theorem	Time Domain $f(t)$	Frequency Domain $F(s)$
Linearity	$a_1 f_1(t) + a_2 f_2(t)$	$a_1 F_1(s) + a_2 F_2(s)$
First Shifting (Frequency)	$e^{-at} f(t)$	$F(s+a)$
Second Shifting (Time)	$f(t-t_0)\, u(t-t_0)$	$e^{-st_0} F(s) \quad (t_0 \geq 0)$
Time Differentiation (1st)	$\dfrac{df(t)}{dt}$	$sF(s) - f(0^-)$
Time Differentiation (2nd)	$\dfrac{d^2 f(t)}{dt^2}$	$s^2 F(s) - sf(0^-) - f'(0^-)$
Time Integration	$\displaystyle\int_{0^-}^{t} f(\tau)\, d\tau$	$\dfrac{F(s)}{s}$
Frequency Differentiation	$t\, f(t)$	$-\dfrac{d}{ds} F(s)$
Time Convolution	$f_1(t) * f_2(t)$	$F_1(s) \cdot F_2(s)$
Initial Value Theorem	$\lim\limits_{t \to 0^+} f(t)$	$\lim\limits_{s \to \infty} sF(s)$ (if limit exists)
Final Value Theorem	$\lim\limits_{t \to \infty} f(t)$	$\lim\limits_{s \to 0} sF(s)$ (poles of $sF(s)$ in LHP)
Periodic Function (T)	$f(t) = f(t+T)$	$\dfrac{1}{1-e^{-sT}} \displaystyle\int_0^T f(t) e^{-st}\, dt$

Critical Examination Note: Validity of the Final Value Theorem (FVT)

The Final Value Theorem $\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ is valid **if and only if** the function $f(t)$ possesses a finite, stable steady-state limit as $t \to \infty$. Mathematically, this requires that all poles of $sF(s)$ reside strictly in the open left-half of the s-plane ($\text{Re}(s) < 0$).

- For $f(t) = \sin(\omega_0 t)$, $F(s) = \frac{\omega_0}{s^2+\omega_0^2}$. Here $sF(s) = \frac{s\omega_0}{s^2+\omega_0^2}$. Evaluating $\lim_{s\to 0} sF(s) = 0$, but $\lim_{t\to\infty} \sin(\omega_0 t)$ does not exist (it oscillates indefinitely). FVT fails because poles are on the $j\omega$-axis!

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2.3 Standard Laplace Transform Pairs

Signal $f(t)$	$F(s)$	Signal $f(t)$	$F(s)$
$\delta(t)$	1	$u(t)$	$\frac{1}{s}$
$t\,u(t)$	$\frac{1}{s^2}$	$t^n\,u(t)$	$\frac{n!}{s^{n+1}}$
$e^{-at}\,u(t)$	$\frac{1}{s+a}$	$t\,e^{-at}\,u(t)$	$\frac{1}{(s+a)^2}$
$\sin(\omega t)\,u(t)$	$\frac{\omega}{s^2+\omega^2}$	$\cos(\omega t)\,u(t)$	$\frac{s}{s^2+\omega^2}$
$e^{-at}\sin(\omega t)\,u(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	$e^{-at}\cos(\omega t)\,u(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$

3. Inverse Laplace Transform and Partial Fraction Expansion

In network transient problems, the s-domain response is generally obtained as a rational transfer function $F(s) = \frac{P(s)}{Q(s)}$, where $P(s)$ and $Q(s)$ are polynomials in s with real coefficients, and $\deg(P) < \deg(Q)$ (strictly proper).

3.1 Decomposition Methodologies

Three Cases of Partial Fraction Expansion

1. **Case I: Distinct Real Roots (Simple Poles)**

$$F(s) = \frac{P(s)}{(s-p_1)(s-p_2)\cdots(s-p_n)} = \sum_{k=1}^{n} \frac{A_k}{s-p_k} \tag{65}$$

The residues A_k are evaluated by Heaviside's cover-up rule:

$$A_k = \lim_{s\to p_k} (s-p_k)\,F(s) \implies f(t) = \sum_{k=1}^{n} A_k\, e^{p_k t}\, u(t) \tag{66}$$

2. **Case II: Repeated Real Roots (Multiple Poles of Order m)**

$$F(s) = \frac{P(s)}{(s-p)^m\, Q_1(s)} = \frac{B_m}{(s-p)^m} + \frac{B_{m-1}}{(s-p)^{m-1}} + \cdots + \frac{B_1}{s-p} + \cdots \tag{67}$$

where the coefficients are given by:

$$B_{m-k} = \frac{1}{k!} \lim_{s\to p} \frac{d^k}{ds^k} \left[(s-p)^m\, F(s)\right], \quad k = 0, 1, \ldots, m-1 \tag{68}$$

3. **Case III: Complex Conjugate Poles ($s = -\alpha \pm j\beta$)**

$$F(s) = \frac{Cs + D}{(s+\alpha)^2 + \beta^2} = \frac{C(s+\alpha) + (D - C\alpha)}{(s+\alpha)^2 + \beta^2} \tag{69}$$

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The inverse transform yields a damped sinusoid:

$$f(t) = Ce^{-\alpha t}\cos(\beta t) + \frac{D - C\alpha}{\beta}e^{-\alpha t}\sin(\beta t) = Me^{-\alpha t}\cos(\beta t - \theta) \tag{70}$$

where $M = \sqrt{C^2 + \left(\frac{D-C\alpha}{\beta}\right)^2}$ and $\theta = \tan^{-1}\left(\frac{D-C\alpha}{\beta C}\right)$.

4. Evaluation of Initial Conditions in Electrical Networks

Evaluating initial conditions at the switching instant ($t = 0^-$ and $t = 0^+$) is an absolute prerequisite for setting up transient differential equations and their Laplace counterparts.

4.1 Physical Boundary and Continuity Principles

Energy Continuity Postulates

1. **Conservation of Inductor Flux Linkage:** An inductor stores magnetic energy $W_m = \frac{1}{2}Li_L^2$. An instantaneous discontinuity in $i_L(t)$ requires infinite voltage ($v_L = L\frac{di}{dt} \to \infty$) and infinite instantaneous power, which is physically impossible in passive circuits without impulse voltage sources. Therefore:

$$i_L(0^+) = i_L(0^-) \tag{71}$$

2. **Conservation of Capacitor Electric Charge:** A capacitor stores electrostatic energy $W_e = \frac{1}{2}Cv_C^2$. An instantaneous discontinuity in $v_C(t)$ requires infinite current ($i_C = C\frac{dv_C}{dt} \to \infty$), which is physically impossible in passive circuits without impulse current sources. Therefore:

$$v_C(0^+) = v_C(0^-) \tag{72}$$

4.2 Equivalent Circuits at $t = 0^-$ and $t = 0^+$

Element	**DC Steady State** ($t = 0^-$)	**At $t = 0^+$ (With Initial State)**	**At $t = 0^+$ (Zero Initial State)**
Resistor (R)	**Resistor** R ($v_R = i_R R$)	**Resistor** R ($v_R = i_R R$)	Resistor R ($v_R = i_R R$)
Inductor (L)	**Short Circuit** ($v_L = 0$)	**Current Source** $I_0 = i_L(0^-)$	Open Circuit ($i_L(0^+) = 0$)
Capacitor (C)	**Open Circuit** ($i_C = 0$)	**Voltage Source** $V_0 = v_C(0^-)$	Short Circuit ($v_C(0^+) = 0$)

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4.3 Evaluation of Higher Derivatives at $t = 0^+$

To evaluate higher-order derivatives such as $\frac{di}{dt}(0^+)$, $\frac{dv_C}{dt}(0^+)$, and $\frac{d^2i}{dt^2}(0^+)$:

1. Write the loop KVL and nodal KCL equations for $t > 0$ containing derivative terms:

$$v_L(t) = L\,\frac{di_L(t)}{dt} \implies \left.\frac{di_L}{dt}\right|_{t=0^+} = \frac{v_L(0^+)}{L} \tag{73}$$

$$i_C(t) = C\,\frac{dv_C(t)}{dt} \implies \left.\frac{dv_C}{dt}\right|_{t=0^+} = \frac{i_C(0^+)}{C} \tag{74}$$

2. Calculate $v_L(0^+)$ and $i_C(0^+)$ directly by evaluating the network loop/node equations at the instant $t = 0^+$ using the known boundary values $i_L(0^+)$ and $v_C(0^+)$.

3. Differentiate the general differential equation with respect to t, and evaluate the resulting relation at $t = 0^+$ to find the second derivative $\frac{d^2i}{dt^2}(0^+)$.

Worked Example: Complete Evaluation of Initial Boundary Derivatives

Problem: In a series RLC circuit ($R = 10\,\Omega, L = 1\,\text{H}, C = 0.1\,\text{F}$), a DC voltage $V = 100\,\text{V}$ is suddenly applied at $t = 0$ by closing a switch. The circuit had no initial energy ($i_L(0^-) = 0, v_C(0^-) = 0$). Find $i(0^+)$, $\frac{di}{dt}(0^+)$, and $\frac{d^2i}{dt^2}(0^+)$.
Solution:

1. **At $t = 0^+$:** Due to continuity of inductor current, $i(0^+) = i_L(0^-) = 0\,\text{A}$. Due to continuity of capacitor voltage, $v_C(0^+) = v_C(0^-) = 0\,\text{V}$.

2. **First derivative $\frac{di}{dt}(0^+)$:** Applying KVL around the loop for $t \geq 0$:

$$R\,i(t) + L\,\frac{di(t)}{dt} + v_C(t) = V \tag{75}$$

Evaluating this at $t = 0^+$:

$$R\,i(0^+) + L\left.\frac{di}{dt}\right|_{t=0^+} + v_C(0^+) = V \implies 10(0) + 1\cdot\left.\frac{di}{dt}\right|_{t=0^+} + 0 = 100 \tag{76}$$

$$\left.\frac{di}{dt}\right|_{t=0^+} = \frac{100 - 0 - 0}{1} = 100\ \text{A/s} \tag{77}$$

3. **Second derivative $\frac{d^2i}{dt^2}(0^+)$:** Differentiating the KVL equation with respect to time t:

$$R\,\frac{di(t)}{dt} + L\,\frac{d^2i(t)}{dt^2} + \frac{dv_C(t)}{dt} = 0 \implies R\,\frac{di}{dt} + L\,\frac{d^2i}{dt^2} + \frac{i(t)}{C} = 0 \tag{78}$$

Evaluating at $t = 0^+$:

$$10(100) + 1\cdot\left.\frac{d^2i}{dt^2}\right|_{t=0^+} + \frac{0}{0.1} = 0 \implies \left.\frac{d^2i}{dt^2}\right|_{t=0^+} = -1000\ \text{A/s}^2 \tag{79}$$

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5. Transformed s-Domain Circuit Models

Instead of formulating differential equations in the time domain and then taking the transform, we can directly map each physical circuit element into its transformed algebraic s-domain model containing initial conditions.

5.1 Derivation of Element Equivalent Circuits

Mathematical Derivation of s-Domain Models

1. **Resistor (R):**

$$v(t) = R\,i(t) \iff V(s) = R\,I(s) \qquad \text{Impedance: } Z_R(s) = R \tag{80}$$

2. **Inductor (L):**

$$v_L(t) = L\,\frac{di_L}{dt} \iff V_L(s) = sL\,I_L(s) - L\,i_L(0^-) \tag{81}$$

- **Thevenin Form (Series):** An impedance sL in series with an opposing voltage source of magnitude $L\,i_L(0^-)$ (positive terminal at the current entry).
- **Norton Form (Parallel):** Solving for $I_L(s)$:

$$I_L(s) = \frac{V_L(s)}{sL} + \frac{i_L(0^-)}{s} \tag{82}$$

An admittance $\frac{1}{sL}$ in parallel with an independent current source $\frac{i_L(0^-)}{s}$ pointing in the direction of $i_L(t)$.

3. **Capacitor (C):**

$$i_C(t) = C\,\frac{dv_C}{dt} \iff I_C(s) = sC\,V_C(s) - C\,v_C(0^-) \tag{83}$$

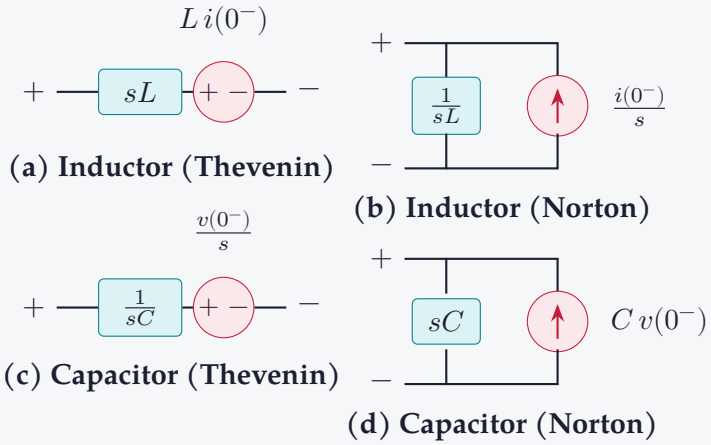
- **Norton Form (Parallel):** An admittance sC in parallel with an independent current source $C\,v_C(0^-)$ directed towards the positive terminal.
- **Thevenin Form (Series):** Solving for $V_C(s)$:

$$V_C(s) = \frac{I_C(s)}{sC} + \frac{v_C(0^-)}{s} \tag{84}$$

An impedance $\frac{1}{sC}$ in series with a voltage source $\frac{v_C(0^-)}{s}$ maintaining the polarity of $v_C(0^-)$.

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TikZ Schematic: Transformed s-Domain Circuit Equivalents with Initial Energy



(a) Inductor (Thevenin) (b) Inductor (Norton)

(c) Capacitor (Thevenin) (d) Capacitor (Norton)

6. Transient Analysis of First-Order Networks (RL and RC)

First-order circuits contain a single equivalent energy-storage element (L or C) and are mathematically governed by linear first-order ordinary differential equations.

6.1 Series RL Network with DC Step Voltage

Consider a series connection of resistance R and inductance L excited at $t = 0$ by a step voltage $V\,u(t)$ with an initial current $i(0^-) = I_0$.

Mathematical Derivation of RL Response

Applying KVL in the transformed s-domain:

$$\frac{V}{s} = R\,I(s) + sL\,I(s) - L\,I_0 \implies I(s)(R+sL) = \frac{V}{s} + L\,I_0 \tag{85}$$

Solving for the current transform $I(s)$:

$$I(s) = \frac{V}{s(sL+R)} + \frac{L\,I_0}{sL+R} = \frac{V/L}{s\left(s+\frac{R}{L}\right)} + \frac{I_0}{s+\frac{R}{L}} \tag{86}$$

Decomposing the first term via partial fractions:

$$\frac{V/L}{s\left(s+\frac{R}{L}\right)} = \frac{V/R}{s} - \frac{V/R}{s+\frac{R}{L}} \tag{87}$$

Substituting back and collecting terms:

$$I(s) = \frac{V/R}{s} - \frac{V/R}{s+\frac{R}{L}} + \frac{I_0}{s+\frac{R}{L}} \tag{88}$$

Taking the inverse Laplace transform for $t \geq 0$:

$$i(t) = \frac{V}{R}\left(1 - e^{-\frac{R}{L}t}\right) + I_0\,e^{-\frac{R}{L}t} = \underbrace{\frac{V}{R}}_{\text{Forced (SS)}} + \underbrace{\left(I_0 - \frac{V}{R}\right)e^{-t/\tau}}_{\text{Natural (Transient)}} \tag{89}$$

where $\tau = \frac{L}{R}$ is the inductive time constant (in seconds).

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Decomposition into Zero-Input and Zero-State Responses

- **Zero-Input Response (ZIR):** Response due entirely to initial internal stored energy ($V = 0$):

$$i_{\text{ZIR}}(t) = I_0\, e^{-t/\tau} \tag{90}$$

- **Zero-State Response (ZSR):** Response due entirely to external applied excitation with zero initial state ($I_0 = 0$):

$$i_{\text{ZSR}}(t) = \frac{V}{R}\left(1 - e^{-t/\tau}\right) \tag{91}$$

- **Principle of Linearity:** Total Response = ZIR + ZSR ≡ Natural + Forced.

6.2 Series RC Network with DC Step Voltage

For a series RC circuit excited by $V\,u(t)$ with initial capacitor voltage $v_C(0^-) = V_0$:

$$\frac{V}{s} = R\,I(s) + \frac{I(s)}{sC} + \frac{V_0}{s} \implies I(s)\left(R + \frac{1}{sC}\right) = \frac{V - V_0}{s} \tag{92}$$

Solving for loop current transform $I(s)$ and capacitor voltage transform $V_C(s)$:

$$I(s) = \frac{V - V_0}{R\left(s + \frac{1}{RC}\right)} \implies i(t) = \frac{V - V_0}{R}\, e^{-t/\tau} \quad (t \geq 0) \tag{93}$$

$$V_C(s) = \frac{V/RC}{s\left(s + \frac{1}{RC}\right)} + \frac{V_0}{s + \frac{1}{RC}} \implies v_C(t) = V\left(1 - e^{-t/\tau}\right) + V_0\, e^{-t/\tau} \tag{94}$$

where $\tau = RC$ is the capacitive time constant.

7. Transient Analysis of Second-Order Networks (RLC)

Second-order circuits contain two uncoupled energy storage elements (L and C) whose transient response is described by a second-order linear differential equation.

7.1 Series RLC Circuit Formulation

Applying KVL to a series RLC circuit excited by a DC voltage $V\,u(t)$ with zero initial conditions:

$$R\,i(t) + L\,\frac{di(t)}{dt} + \frac{1}{C}\int_0^t i(\tau)\,d\tau = V \tag{95}$$

Taking the Laplace transform with zero initial conditions:

$$I(s)\left[R + sL + \frac{1}{sC}\right] = \frac{V}{s} \implies I(s)\left[\frac{s^2L + sR + \frac{1}{C}}{s}\right] = \frac{V}{s} \tag{96}$$

$$I(s) = \frac{V/L}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = \frac{V/L}{s^2 + 2\alpha s + \omega_0^2} \tag{97}$$

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Standard Second-Order Characteristic Parameters

$$s^2 + 2\alpha s + \omega_0^2 = 0 \quad \text{or} \quad s^2 + 2\zeta\omega_0 s + \omega_0^2 = 0 \tag{98}$$

- **Neper Frequency (Attenuation Factor):** $\alpha = \dfrac{R}{2L}$ (rad/s)
- **Undamped Resonant Frequency:** $\omega_0 = \dfrac{1}{\sqrt{LC}}$ (rad/s)
- **Damping Ratio:** $\zeta = \dfrac{\alpha}{\omega_0} = \dfrac{R}{2}\sqrt{\dfrac{C}{L}}$

Roots of the characteristic polynomial:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} = -\zeta\omega_0 \pm \omega_0\sqrt{\zeta^2 - 1} \tag{99}$$

7.2 Classification of Damping Regimes

The nature of the transient response depends strictly on the discriminant $(\alpha^2 - \omega_0^2)$ or damping ratio ζ:

Three Classical Damping Regimes

1. **Overdamped** ($\zeta > 1 \iff R > 2\sqrt{L/C} \iff \alpha > \omega_0$)**:** The roots s_1, s_2 are real, negative, and unequal.

$$i(t) = \frac{V}{2\beta L}\left(e^{-(\alpha-\beta)t} - e^{-(\alpha+\beta)t}\right), \quad \text{where } \beta = \sqrt{\alpha^2 - \omega_0^2} \tag{100}$$

   The current rises to a peak and decays sluggishly to zero without any oscillation.

2. **Critically Damped** ($\zeta = 1 \iff R = 2\sqrt{L/C} \iff \alpha = \omega_0$)**:** The roots are real, negative, and identical: $s_1 = s_2 = -\alpha$.

$$i(t) = \frac{V}{L}\, t\, e^{-\alpha t} \tag{101}$$

   Provides the fastest non-oscillatory return to steady-state equilibrium.

3. **Underdamped** ($\zeta < 1 \iff R < 2\sqrt{L/C} \iff \alpha < \omega_0$)**:** The roots are a pair of complex conjugates in the left-half s-plane:

$$s_{1,2} = -\alpha \pm j\omega_d, \quad \text{where } \omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0\sqrt{1 - \zeta^2} \tag{102}$$

   ω_d is the damped natural frequency. The loop current is an exponentially decaying sinusoidal oscillation:

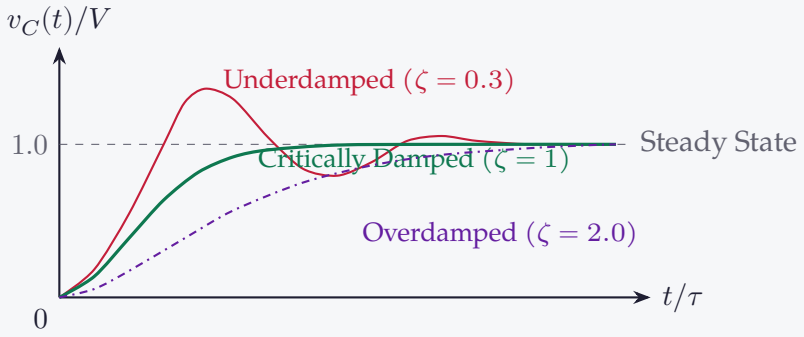
$$i(t) = \frac{V}{\omega_d L}\, e^{-\alpha t} \sin(\omega_d t) \tag{103}$$

   The capacitor voltage exhibits oscillatory overshoot:

$$v_C(t) = V\left[1 - \frac{e^{-\alpha t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \phi)\right], \quad \phi = \cos^{-1}(\zeta) \tag{104}$$

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TikZ Plot: Comparison of Second-Order Step Responses for Different Damping Ratios



7.3 Dual Parallel RLC Network

In a parallel RLC circuit excited by an independent current step $I\,u(t)$, the dual nodal formulation gives:

$$\frac{v(t)}{R} + \frac{1}{L}\int_0^t v(\tau)\,d\tau + C\,\frac{dv(t)}{dt} = I \tag{105}$$

The characteristic equation and parameters are:

$$s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0 \implies \alpha = \frac{1}{2RC}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{1}{2R}\sqrt{\frac{L}{C}} \tag{106}$$

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Quick Revision — Module 3

Laplace Transforms, Network Transients and Initial Conditions | EC304

Key Formulas and Network Relations at a Glance

Unilateral Laplace Transform	$F(s) = \int_{0^-}^{\infty} f(t)\, e^{-st}\, dt, \quad s = \sigma + j\omega$
Time Derivative Transform	$\mathcal{L}\{f'(t)\} = sF(s) - f(0^-), \quad \mathcal{L}\{f''(t)\} = s^2F(s) - sf(0^-) - f'(0^-)$
Time Shifting Property	$\mathcal{L}\{f(t-t_0)u(t-t_0)\} = e^{-st_0}F(s) \quad (t_0 \geq 0)$
Final Value Theorem	$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ (poles in open LHP)
Inductor s-Domain Model	$V_L(s) = sLI_L(s) - Li_L(0^-) \iff I_L(s) = \frac{V_L(s)}{sL} + \frac{i_L(0^-)}{s}$
Capacitor s-Domain Model	$V_C(s) = \frac{I_C(s)}{sC} + \frac{v_C(0^-)}{s} \iff I_C(s) = sCV_C(s) - Cv_C(0^-)$
First-Order Step Response	$x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}, \quad \tau_{RL} = \frac{L}{R}, \ \tau_{RC} = RC$
Series RLC Damping Parameters	$\alpha = \frac{R}{2L}, \quad \omega_0 = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{\alpha}{\omega_0} = \frac{R}{2}\sqrt{\frac{C}{L}}$
Damped Frequency	$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \omega_0\sqrt{1-\zeta^2} \quad (\text{for } \zeta < 1)$

One-Line Definitions

- **Singularity Function:** A function exhibiting discontinuities in its value or its higher-order time derivatives.
- **Unit Impulse $\delta(t)$:** A generalized function of infinite height, zero width, and unit integral area at $t = 0$.
- **Lower Limit 0^-:** The instant immediately prior to switching, capturing jump discontinuities and initial states.
- **Zero-Input Response (ZIR):** System response produced solely by initial stored energy with external sources zeroed.
- **Zero-State Response (ZSR):** System response produced solely by applied excitations with zero initial stored energy.
- **Time Constant (τ):** Time required for a first-order transient response to decay to 36.8% $(1/e)$ of initial value.
- **Neper Frequency (α):** Exponential damping attenuation factor governing the rate of response decay.
- **Undamped Resonant Frequency (ω_0):** Natural oscillation frequency of an LC circuit in the absence of resistance.
- **Damping Ratio (ζ):** Dimensionless ratio α/ω_0 determining whether a second-order circuit oscillates.
- **Critically Damped Circuit:** Second-order circuit with $\zeta = 1$ returning to steady state in minimum non-oscillatory time.

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Frequently Asked / Viva Questions

Q1. Why cannot inductor current change instantaneously in a finite circuit?

Stored magnetic energy is $W = \frac{1}{2}Li_L^2$. An instantaneous change requires infinite power ($P = \frac{dW}{dt} \to \infty$) and infinite voltage ($v_L = L\frac{di}{dt} \to \infty$), which is impossible with finite sources. Hence, $i_L(0^+) = i_L(0^-)$.

Q2. Why does an uncharged capacitor act as a short circuit at $t = 0^+$?

Since $v_C(0^+) = v_C(0^-) = 0\,\text{V}$, the terminal voltage is zero immediately after switching while current flows freely. An element supporting finite current with zero voltage drop behaves identically to an ideal short circuit.

Q3. Explain the physical necessity of the lower limit 0^- in unilateral Laplace transforms.

The 0^- limit captures Dirac delta impulses $\delta(t)$ acting at $t = 0$ and accounts for pre-switching initial conditions ($i_L(0^-), v_C(0^-)$). A lower limit of 0^+ would omit impulses occurring at the switching origin.

Q4. State the validity condition for applying the Final Value Theorem.

$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$ is valid strictly if all poles of $sF(s)$ lie in the open left-half s-plane. If poles lie on the imaginary axis (e.g. sustained sinusoids) or in the right-half plane, FVT produces incorrect results.

Q5. Distinguish between natural and forced responses in circuit transients.

The natural (transient) response depends solely on circuit storage elements (R, L, C) and decays to zero. The forced (steady-state) response is sustained by external driving sources and mirrors their functional form.

Q6. Under what switching condition is an AC RL circuit completely transient-free?

In an RL circuit excited by $v(t) = V_m \sin(\omega t + \theta)$, transient-free operation occurs when switching at $\theta - \phi = 0$ or π (where $\phi = \tan^{-1}\frac{\omega L}{R}$), matching prospective steady-state current to the initial zero current.

Q7. How does the Norton s-domain model of an inductor differ from its Thevenin form?

The Thevenin form consists of an impedance sL in series with an opposing voltage source $Li_L(0^-)$. The Norton form comprises an admittance $\frac{1}{sL}$ in parallel with a current source $\frac{i_L(0^-)}{s}$ pointing in the direction of i_L.

Q8. Define damping ratio ζ and give its formulas for series and parallel RLC circuits.

$\zeta = \alpha/\omega_0$. For series RLC: $\zeta = \frac{R}{2}\sqrt{\frac{C}{L}}$. For parallel RLC: $\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$.

Q9. Why is critical damping ($\zeta = 1$) preferred in analog meters and galvanometers?

Critical damping ensures the pointer reaches its steady deflection in the shortest possible time without oscillatory overshoot or sluggish delay, enabling fast, accurate readings.

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Frequently Asked / Viva Questions (contd.)

Q10. What is the circuit significance of the unit doublet $\delta'(t)$?

The unit doublet represents $\frac{d\delta}{dt}$. In circuits, it arises from impulsive current injection across inductors or impulse voltage across capacitors, abruptly altering state variables at $t = 0$.

Summary Matrix: Comprehensive Comparison of First- and Second-Order Transients

Circuit Type	Time Const. / Damping	Roots	Transient Response Nature
Series RL	$\tau = \frac{L}{R}$	$s = -\frac{R}{L}$	Monotonic exponential rise/decay; no oscillation.
Series RC	$\tau = RC$	$s = -\frac{1}{RC}$	Monotonic exponential voltage change; current step at $t = 0^+$.
Series RLC Overdamped $\zeta > 1$	$\alpha = \frac{R}{2L} > \omega_0$, $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	**Sum of two decaying exponentials; sluggish, non-oscillatory.**	
Series RLC Critically Damped $\zeta = 1$	$\alpha = \omega_0$, $s_1 = s_2 = -\alpha$	**Rapid monotonic return to equilibrium without overshoot ($(A_1 + A_2 t)e^{-\alpha t}$).**	
Series RLC Underdamped $\zeta < 1$	$\alpha < \omega_0$, $s_{1,2} = -\alpha \pm j\omega_d$	**Damped sinusoidal oscillations with decaying envelope $e^{-\alpha t}$.**	
Parallel RLC $\zeta = \frac{1}{2R}\sqrt{\frac{L}{C}}$	$\alpha = \frac{1}{2RC}$, $s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2}$	**Dual of series RLC; higher R produces lower damping (oscillatory).**	

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Module 4
Complex Frequency and Network Functions

1. Complex Frequency and Network Functions

The concept of complex frequency generalizes real sinusoidal and exponential signals into a unified mathematical framework. It forms the foundation for characterizing linear time-invariant (LTI) networks in the complex frequency s-plane.

1.1 Concept of Complex Frequency

Definition of Complex Frequency

A generalized damped sinusoidal signal is expressed as:

$$f(t) = A\,e^{\sigma t}\cos(\omega t + \theta) = \text{Re}\left\{A\,e^{j\theta}\,e^{(\sigma+j\omega)t}\right\} = \text{Re}\left\{\mathbf{F}\,e^{st}\right\} \tag{107}$$

The parameter $s = \sigma + j\omega$ is the **complex frequency** (measured in s^{-1} or complex nepers/radians per second):

- $\sigma = \text{Re}(s)$ is the **Neper frequency** (damping or growth rate in Np/s).
- $\omega = \text{Im}(s)$ is the **radian angular frequency** (oscillation rate in rad/s).

Special Cases of the Complex Frequency Variable $s = \sigma + j\omega$

1. $\sigma = 0, \omega = 0 \implies s = 0$: Constant DC signal ($f(t) = A$).
2. $\sigma \neq 0, \omega = 0 \implies s = \sigma$: Pure exponential signal ($e^{\sigma t}$; decaying if $\sigma < 0$, growing if $\sigma > 0$).
3. $\sigma = 0, \omega \neq 0 \implies s = j\omega$: Undamped steady-state sinusoidal signal ($A\cos(\omega t)$).
4. $\sigma < 0, \omega \neq 0$: Damped sinusoidal signal (exponentially decaying oscillatory transient).
5. $\sigma > 0, \omega \neq 0$: Growing sinusoidal signal (unstable active system).

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1.2 Definition and Classification of Network Functions

Network Function Definition

For a linear, lumped, time-invariant network with all initial conditions set to zero (relaxed state), a **Network Function** $H(s)$ is defined as the ratio of the Laplace transform of the output response $R(s)$ to the Laplace transform of the input excitation $E(s)$:

$$H(s) = \frac{R(s)}{E(s)} \iff R(s) = H(s) \cdot E(s) \tag{108}$$

Network functions are broadly classified into two categories:

1. **Driving-Point Immittance Functions:** Excitation and response are measured at the **same port**:
   - *Driving-point impedance:* $Z_{in}(s) = \dfrac{V_1(s)}{I_1(s)}$
   - *Driving-point admittance:* $Y_{in}(s) = \dfrac{I_1(s)}{V_1(s)} = \dfrac{1}{Z_{in}(s)}$
2. **Transfer Functions:** Excitation and response are measured at **different ports**:
   - *Voltage Transfer Function (Gain):* $G_{21}(s) = \dfrac{V_2(s)}{V_1(s)}$
   - *Current Transfer Function (Gain):* $\alpha_{21}(s) = \dfrac{I_2(s)}{I_1(s)}$
   - *Transfer Impedance:* $Z_{21}(s) = \dfrac{V_2(s)}{I_1(s)}$
   - *Transfer Admittance:* $Y_{21}(s) = \dfrac{I_2(s)}{V_1(s)}$

2. Poles and Zeros of Immittance Functions and their Properties

Any network function of a lumped LTI circuit can be expressed as a rational function of s with real coefficients:

$$H(s) = \frac{N(s)}{D(s)} = \frac{a_m s^m + a_{m-1}s^{m-1} + \cdots + a_1 s + a_0}{b_n s^n + b_{n-1}s^{n-1} + \cdots + b_1 s + b_0} = K\frac{\prod_{i=1}^{m}(s - z_i)}{\prod_{k=1}^{n}(s - p_k)} \tag{109}$$

2.1 Definitions of Poles and Zeros

Poles and Zeros

- **Zeros (z_i):** The values of complex frequency s for which $H(s) = 0$ (roots of the numerator $N(s) = 0$). At a transmission zero, the network completely suppresses the input signal.
- **Poles (p_k):** The values of complex frequency s for which $H(s) \to \infty$ (roots of the

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denominator $D(s) = 0$). Poles define the natural frequencies (modes) and determine stability.

- **Critical Frequencies at Infinity:** If $n > m$, $H(s)$ has $(n-m)$ zeros at $s = \infty$. If $m > n$, $H(s)$ has $(m-n)$ poles at $s = \infty$. The total number of zeros equals the total number of poles in the extended complex plane.

2.2 Properties of Driving-Point Immittances (Positive Real Functions)

For any passive, bilateral electrical network without internal active sources, the driving-point immittance $Z(s)$ or $Y(s)$ must be a **Positive Real (PR) Function** (Brune's condition):

$$\text{Re}\{Z(s)\} \geq 0 \quad \text{whenever} \quad \text{Re}(s) \geq 0 \tag{110}$$

Necessary and Sufficient Properties of Passive Immittance Functions

1. **Hurwitz Denominator and Numerator:** All poles and zeros of $Z(s)$ must lie in the left half of the s-plane ($\text{Re}(s) \leq 0$). They cannot reside in the open right-half plane (RHP).
2. **Imaginary Axis Poles and Zeros:** Any poles and zeros located on the $j\omega$-axis must be **simple** (multiplicity of 1). Multiple poles on the $j\omega$-axis are strictly forbidden because they generate unbounded polynomial-exponential terms ($t \sin \omega t$).
3. **Residue Condition:** The residues of $Z(s)$ at simple poles on the $j\omega$-axis must be **real and strictly positive**.
4. **Degree Constraint:** The degrees of the numerator polynomial $N(s)$ and denominator polynomial $D(s)$ can differ at most by 1: $|m-n| \leq 1$.
5. **Lowest Degree Constraint:** The lowest power of s in $N(s)$ and $D(s)$ can differ at most by 1.
6. **Foster Alternation Law (LC Immittance):** For purely reactive lossless LC networks, all poles and zeros lie exclusively on the $j\omega$-axis, are simple, and must strictly **interlace (alternate)**:
$$0 \leq \omega_{z1} < \omega_{p1} < \omega_{z2} < \omega_{p2} < \cdots \tag{111}$$
Furthermore, the slope of reactance with respect to frequency is strictly positive: $\frac{dX(\omega)}{d\omega} > 0$.

3. Sinusoidal Steady-State Response from Pole-Zero Locations

The steady-state frequency response $H(j\omega)$ of any network function can be evaluated directly by geometric inspection of its pole-zero constellation in the s-plane.

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3.1 Geometric Vector Formulation

Evaluating $H(s)$ on the imaginary axis $s = j\omega$:

$$H(j\omega) = K\frac{(j\omega - z_1)(j\omega - z_2)\cdots(j\omega - z_m)}{(j\omega - p_1)(j\omega - p_2)\cdots(j\omega - p_n)} \tag{112}$$

Each term $(j\omega - z_i)$ represents a directed complex vector in the s-plane from the zero z_i to the test frequency point $s = j\omega$:

$$(j\omega - z_i) = M_i\, e^{j\psi_i}, \quad \text{where } M_i = |j\omega - z_i|, \quad \psi_i = \angle(j\omega - z_i) \tag{113}$$

Similarly, each term $(j\omega - p_k)$ represents a vector from the pole p_k to the test point $j\omega$:

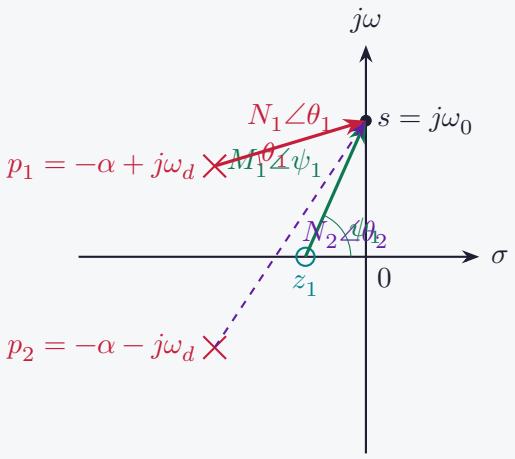
$$(j\omega - p_k) = N_k\, e^{j\theta_k}, \quad \text{where } N_k = |j\omega - p_k|, \quad \theta_k = \angle(j\omega - p_k) \tag{114}$$

Magnitude and Phase Computation via Pole-Zero Geometry

$$|H(j\omega)| = |K| \cdot \frac{M_1 M_2 \cdots M_m}{N_1 N_2 \cdots N_n} = |K| \cdot \frac{\prod(\text{distances from zeros to } j\omega)}{\prod(\text{distances from poles to } j\omega)} \tag{115}$$

$$\angle H(j\omega) = \angle K + \sum_{i=1}^{m}\psi_i - \sum_{k=1}^{n}\theta_k = \angle K + \sum(\text{zero angles}) - \sum(\text{pole angles}) \tag{116}$$

TikZ Schematic: Geometric Vector Evaluation in the Complex s-Plane



4. The Convolution Theorem in Electrical Networks

The convolution theorem establishes the fundamental mathematical bridge between time-domain input-output relationships and s-domain transfer function multiplication.

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4.1 Mathematical Definition and Circuit Interpretation

Time Convolution Theorem

Let $h(t)$ be the **impulse response** of a relaxed LTI network (the response when excited by $x(t) = \delta(t)$). The zero-state response $y(t)$ to an arbitrary continuous excitation $x(t)$ applied at $t = 0$ is given by the convolution integral:

$$y(t) = x(t) * h(t) = \int_{0^-}^{t} x(\tau)\, h(t-\tau)\, d\tau = \int_{0^-}^{t} h(\tau)\, x(t-\tau)\, d\tau \tag{117}$$

In the transformed frequency domain:

$$\mathcal{L}\{x(t) * h(t)\} = X(s) \cdot H(s) \iff Y(s) = H(s)X(s) \tag{118}$$

where $H(s) = \mathcal{L}\{h(t)\}$ is the transfer function of the network.

Relation Between Impulse Response and Step Response

If $s(t)$ represents the unit step response ($x(t) = u(t)$) of a linear network:

$$s(t) = u(t) * h(t) = \int_{0^-}^{t} h(\tau)\, d\tau \iff h(t) = \frac{ds(t)}{dt} \tag{119}$$

The unit impulse response is the exact time derivative of the unit step response.

5. Two-Port Network Parameters and Interconnections

A two-port network is a four-terminal electrical network having two distinct access ports: Port 1 (input port, terminals $1-1'$) and Port 2 (output port, terminals $2-2'$). Standard sign convention assumes terminal currents I_1 and I_2 enter the positive terminal of their respective ports.

TikZ Schematic: Standard Terminal Variable Convention for a Two-Port Network



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5.1 The Six Fundamental Two-Port Parameter Sets

Definitions and Equations of Two-Port Parameter Sets

1. **Impedance (Z) Parameters (Open-Circuit Parameters):**

$$\begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \implies \begin{aligned} V_1 &= z_{11}I_1 + z_{12}I_2 \\ V_2 &= z_{21}I_1 + z_{22}I_2 \end{aligned} \tag{120}$$

$z_{11} = \frac{V_1}{I_1}\big|_{I_2=0}$ (input impedance), $z_{12} = \frac{V_1}{I_2}\big|_{I_1=0}$ (reverse transfer impedance),
$z_{21} = \frac{V_2}{I_1}\big|_{I_2=0}$ (forward transfer impedance), $z_{22} = \frac{V_2}{I_2}\big|_{I_1=0}$ (output impedance).

2. **Admittance (Y) Parameters (Short-Circuit Parameters):**

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} \implies \begin{aligned} I_1 &= y_{11}V_1 + y_{12}V_2 \\ I_2 &= y_{21}V_1 + y_{22}V_2 \end{aligned} \tag{121}$$

$[Y] = [Z]^{-1} \implies y_{11} = \frac{z_{22}}{\Delta z}, \quad y_{12} = -\frac{z_{12}}{\Delta z}, \quad y_{21} = -\frac{z_{21}}{\Delta z}, \quad y_{22} = \frac{z_{11}}{\Delta z}.$

3. **Transmission ($ABCD$) Parameters (Cascade / Chain Parameters):** Relates input terminal variables to output terminal variables under reverse current load convention:

$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix} \implies \begin{aligned} V_1 &= AV_2 - BI_2 \\ I_1 &= CV_2 - DI_2 \end{aligned} \tag{122}$$

$A = \frac{V_1}{V_2}\big|_{I_2=0}$ (reverse voltage ratio), $B = \frac{V_1}{-I_2}\big|_{V_2=0}$ (transfer impedance),
$C = \frac{I_1}{V_2}\big|_{I_2=0}$ (transfer admittance), $D = \frac{I_1}{-I_2}\big|_{V_2=0}$ (reverse current ratio).

4. **Hybrid (h) Parameters:** Used extensively in small-signal bipolar transistor modeling:

$$\begin{bmatrix} V_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ V_2 \end{bmatrix} \implies \begin{aligned} V_1 &= h_{11}I_1 + h_{12}V_2 \\ I_2 &= h_{21}I_1 + h_{22}V_2 \end{aligned} \tag{123}$$

5. **Inverse Hybrid (g) Parameters:** Used in field-effect transistor modeling:

$$\begin{bmatrix} I_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ I_2 \end{bmatrix} \implies \begin{aligned} I_1 &= g_{11}V_1 + g_{12}I_2 \\ V_2 &= g_{21}V_1 + g_{22}I_2 \end{aligned} \tag{124}$$

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5.2 Reciprocity and Symmetry Conditions

Parameter Set	Reciprocity Condition	Symmetry Condition
Impedance (Z)	$z_{12} = z_{21}$	$z_{11} = z_{22}$
Admittance (Y)	$y_{12} = y_{21}$	$y_{11} = y_{22}$
Transmission $(ABCD)$	$AD - BC = 1$	$A = D$
Hybrid (h)	$h_{12} = -h_{21}$	$\Delta h = h_{11}h_{22} - h_{12}h_{21} = 1$
Inverse Hybrid (g)	$g_{12} = -g_{21}$	$\Delta g = g_{11}g_{22} - g_{12}g_{21} = 1$

5.3 Interconnections of Two-Port Networks

When two-port networks N_A and N_B are interconnected, their overall parameter matrices combine algebraically:

Combination Rules for Interconnected Two-Ports

1. **Series-Series Connection:** Input ports connected in series, output ports in series:

$$[Z] = [Z_A] + [Z_B] \tag{125}$$

2. **Parallel-Parallel Connection:** Input ports in parallel, output ports in parallel:

$$[Y] = [Y_A] + [Y_B] \tag{126}$$

3. **Cascade (Tandem) Connection:** Output of N_A connected to input of N_B:

$$[T] = [T_A] \times [T_B] = \begin{bmatrix} A_A & B_A \\ C_A & D_A \end{bmatrix} \begin{bmatrix} A_B & B_B \\ C_B & D_B \end{bmatrix} \tag{127}$$

4. **Series-Parallel Connection:** Input ports in series, output ports in parallel:

$$[h] = [h_A] + [h_B] \tag{128}$$

Brune's Test for Validity: When combining two-port networks in series or parallel, circulating loop currents between the two networks can violate the fundamental port condition ($i_{\text{in}} = i_{\text{return}}$). Brune's test confirms validity: if the terminal voltage across the interconnection points is zero before connection, the matrices can be added directly without ideal isolation transformers.

6. Behaviors of Series and Parallel Resonant Circuits

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Resonance is a phenomenon in reactive circuits where inductive and capacitive reactances cancel each other out at a specific frequency, causing the total input impedance to become purely real and the circuit power factor to equal unity ($\cos\phi = 1$).

6.1 Series Resonance

Consider an RLC branch connected in series across an AC source $\mathbf{V}$. The total complex impedance is:

$$\mathbf{Z}(j\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right) \tag{129}$$

Series Resonance Equations

1. **Resonance Condition:** Net reactance is identically zero:

$$X_L = X_C \implies \omega_0 L = \frac{1}{\omega_0 C} \implies \omega_0 = \frac{1}{\sqrt{LC}} \text{ rad/s}, \quad f_0 = \frac{1}{2\pi\sqrt{LC}} \text{ Hz} \tag{130}$$

2. **Impedance at Resonance:** Minimum and purely resistive:

$$\mathbf{Z}(j\omega_0) = Z_{\min} = R \implies \mathbf{I}_0 = \frac{\mathbf{V}}{R} \quad \text{(Maximum Current)} \tag{131}$$

3. **Quality Factor (Q):** Defined as $2\pi \times \frac{\text{Maximum Energy Stored}}{\text{Energy Dissipated per Cycle}}$:

$$Q_0 = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR} = \frac{1}{R}\sqrt{\frac{L}{C}} \tag{132}$$

4. **Voltage Magnification:** Reactive voltages across L and C can far exceed source voltage:

$$V_L(j\omega_0) = V_C(j\omega_0) = Q_0 \cdot V \tag{133}$$

5. **Half-Power Frequencies and Bandwidth (BW):** The frequencies ω_1 (lower) and ω_2 (upper) where current magnitude drops to $I_0/\sqrt{2}$ (70.7%) and dissipated active power drops to $P_{\max}/2$:

$$\omega_{1,2} = \mp\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}} \implies BW = \Delta\omega = \omega_2 - \omega_1 = \frac{R}{L} = \frac{\omega_0}{Q_0} \tag{134}$$

Geometric center frequency relation: $\omega_0 = \sqrt{\omega_1\omega_2}$. For high-$Q$ circuits ($Q \geq 10$), arithmetic symmetry holds: $\omega_1 \approx \omega_0 - \frac{BW}{2}$, $\omega_2 \approx \omega_0 + \frac{BW}{2}$.

6.2 Parallel Resonance (Practical Tank Circuit)

A practical inductor possesses winding resistance R_L. The practical parallel tank consists of a coil (R_L in series with L) connected in parallel with an ideal capacitor C.

$$\mathbf{Y} = \frac{1}{R_L + j\omega L} + j\omega C = \frac{R_L - j\omega L}{R_L^2 + \omega^2 L^2} + j\omega C = \frac{R_L}{R_L^2 + \omega^2 L^2} + j\left(\omega C - \frac{\omega L}{R_L^2 + \omega^2 L^2}\right) \tag{135}$$

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Practical Tank Circuit Resonance Conditions

1. **Unity Power Factor Resonant Frequency ω_r:** Setting net susceptance to zero:

$$\omega_r C = \frac{\omega_r L}{R_L^2 + \omega_r^2 L^2} \implies \omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}} = \omega_0 \sqrt{1 - \frac{R_L^2 C}{L}} \tag{136}$$

   Physical condition for parallel resonance to occur: $\dfrac{L}{C} > R_L^2$.

2. **Dynamic Impedance (Z_d):** At resonance, the circuit exhibits **maximum impedance** (anti-resonance):

$$Z_d = \frac{1}{Y_{\text{min}}} = \frac{R_L^2 + \omega_r^2 L^2}{R_L} = \frac{L}{CR_L} \quad (\Omega) \tag{137}$$

   Total line current is minimum: $I_{\text{min}} = \frac{V}{Z_d} = \frac{VCR_L}{L}$.

3. **Current Magnification:** Circulating loop current between coil and capacitor is Q times line current:

$$I_{\text{circ}} \approx Q \cdot I_{\text{line}}, \quad \text{where } Q = \frac{\omega_r L}{R_L} \tag{138}$$

7. Introduction to Passive Electric Filters

An electric filter is a frequency-selective two-port network designed to transmit signals within specified frequency bands (**pass bands**) with minimal attenuation, while significantly attenuating signals outside those bands (**attenuation / stop bands**).

7.1 Propagation Characteristics and Cut-Off Frequency

For a symmetrical reactive network terminated in its characteristic impedance Z_0:

$$\frac{\mathbf{V}_1}{\mathbf{V}_2} = \frac{\mathbf{I}_1}{\mathbf{I}_2} = e^{\gamma} = e^{\alpha + j\beta} \implies \gamma = \alpha + j\beta \tag{139}$$

where γ is the complex propagation constant, α is the attenuation constant (in Nepers), and β is the phase shift constant (in radians).

- **Pass Band:** $\alpha = 0$, signals pass with zero attenuation ($|\mathbf{V}_2/\mathbf{V}_1| = 1$), experiencing only a phase delay β.
- **Stop Band:** $\alpha > 0$, signal amplitude decays exponentially.
- **Cut-Off Frequency (f_c):** The boundary frequency demarcating the pass band from the stop band, defined where $\alpha = 0$ begins to increase.

7.2 Constant-k Prototype Filter Sections

A symmetrical ladder network has series impedance Z_1 and shunt impedance Z_2. It is termed a **constant-k filter** if:

$$Z_1 \cdot Z_2 = k^2 = R_0^2 \quad \text{(independent of frequency)} \tag{140}$$

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where $R_0 = \sqrt{L/C}$ is the nominal design characteristic resistance.

Filter Type	Series Arm Z_1	Shunt Arm Z_2	Cut-off & Design Formulas
Low-Pass (LPF) $L = \frac{R_0}{\pi f_c}, \quad C = \frac{1}{\pi f_c R_0}$	**Inductor** L $(j\omega L)$	**Capacitor** C $(\frac{1}{j\omega C})$	$f_c = \frac{1}{\pi\sqrt{LC}}$,
High-Pass (HPF) $L = \frac{R_0}{4\pi f_c}, \quad C = \frac{1}{4\pi f_c R_0}$	**Capacitor** C $(\frac{1}{j\omega C})$	**Inductor** L $(j\omega L)$	$f_c = \frac{1}{4\pi\sqrt{LC}}$,
Band-Pass (BPF)	**Series resonant** (L_1, C_1)	**Parallel resonant** (L_2, C_2)	$f_0 = \sqrt{f_1 f_2}, \quad BW = f_2 - f_1$
Band-Reject (BRF)	**Parallel resonant** (L_1, C_1)	**Series resonant** (L_2, C_2)	Stops band between f_1 and f_2; passes DC and high AC.

Limitations of Constant-k Filters and Evolution to m-Derived Filters

1. **Gradual Attenuation Rise:** The attenuation α of a constant-k filter rises slowly beyond the cut-off frequency f_c.
2. **Mismatched Terminal Impedance:** The characteristic impedance $Z_{0T}(f)$ or $Z_{0\pi}(f)$ varies widely across the pass band and drops to zero or infinity at f_c, creating severe reflection losses when terminated in a fixed resistance R_0.
3. **m-Derived Filter Fix:** By incorporating a tuning parameter $m = \sqrt{1 - (f_c/f_\infty)^2}$ $(0 < m \leq 1)$, an infinite attenuation notch (f_∞) is introduced near the cut-off frequency, producing an extraordinarily sharp transition and a nearly constant characteristic impedance over 85% of the pass band.

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Quick Revision — Module 4

Network Functions, Two-Port Networks, Resonance and Filters | EC304

Key Formulas and Network Relations at a Glance

Complex Frequency	$s = \sigma + j\omega \quad (\sigma : \text{Np/s},\ \omega : \text{rad/s})$
Magnitude from Pole-Zero	$\lvert H(j\omega)\rvert = \lvert K\rvert \frac{\prod \lvert j\omega - z_i\rvert}{\prod \lvert j\omega - p_k\rvert} = \lvert K\rvert \frac{\text{Distance product from zeros}}{\text{Distance product from poles}}$
Phase from Pole-Zero	$\angle H(j\omega) = \sum \angle(j\omega - z_i) - \sum \angle(j\omega - p_k)$
Z **to** Y **Matrix Inverse**	$[Y] = [Z]^{-1}, \quad y_{11} = \frac{z_{22}}{\Delta z},\ y_{12} = -\frac{z_{12}}{\Delta z},\ y_{22} = \frac{z_{11}}{\Delta z}$
$ABCD$ **Transmission Cascade**	$[T_{\text{total}}] = [T_A] \times [T_B], \quad AD - BC = 1$ (Reciprocity)
Series Resonance	$\omega_0 = \frac{1}{\sqrt{LC}}, \quad Z_{\min} = R, \quad Q = \frac{\omega_0 L}{R} = \frac{1}{R}\sqrt{\frac{L}{C}}$
Bandwidth & Selectivity	$BW = \omega_2 - \omega_1 = \frac{R}{L} = \frac{\omega_0}{Q}, \quad \omega_0 = \sqrt{\omega_1 \omega_2}$
Parallel Dynamic Impedance	$\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}}, \quad Z_d = \frac{L}{CR_L}$ (Max Impedance)
Constant-k LPF Cut-off	$f_c = \frac{1}{\pi\sqrt{LC}}, \quad L = \frac{R_0}{\pi f_c}, \quad C = \frac{1}{\pi f_c R_0}$

One-Line Definitions

- **Complex Frequency (s):** Variable $s = \sigma + j\omega$ combining exponential damping rate and oscillation frequency.
- **Driving-Point Immittance:** The input impedance or admittance measured across a single terminal pair with internal relaxed state.
- **Transfer Function:** Ratio of response transform at one port to excitation transform at another port under zero initial states.
- **Hurwitz Polynomial:** A polynomial having all its roots located strictly in the closed left half of the complex s-plane.
- **Positive Real (PR) Function:** A rational function $Z(s)$ whose real part is non-negative whenever the real part of s is non-negative.
- **Impulse Response $h(t)$:** Time-domain output of an initially relaxed network when driven by an ideal Dirac delta excitation.
- **Quality Factor (Q):** Dimensionless figure of merit measuring energy storage efficiency relative to per-cycle dissipation.
- **Anti-Resonance:** Parallel resonance state exhibiting maximum terminal impedance and minimum source line current.
- **Cut-Off Frequency (f_c):** Frequency demarcating pass band ($\alpha = 0$) from stop band ($\alpha > 0$) in a filter network.
- **m-Derived Filter:** Modified ladder section with a resonance trap providing sharp attenuation notch and flat Z_0.

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Frequently Asked / Viva Questions

Q1. What are the necessary conditions for a rational function to be a Positive Real (PR) function?

$Z(s)$ must have real coefficients; all poles and zeros must lie in the left half s-plane; $j\omega$-axis poles must be simple with real, positive residues; and degrees of $N(s)$ and $D(s)$ can differ at most by 1.

Q2. Why must poles and zeros alternate on the imaginary axis for an LC driving-point immittance?

By Foster's Reactance Theorem, $\frac{dX(\omega)}{d\omega} > 0$ strictly holds for lossless LC networks. Reactance increases monotonically with frequency, forcing every zero ($X = 0$) to be succeeded by a pole ($X \to \infty$).

Q3. State the condition for reciprocity and symmetry in terms of $ABCD$ parameters.

Reciprocity holds if the transmission determinant is unity: $AD - BC = 1$. The network is symmetrical (electrically identical when ports are interchanged) if $A = D$.

Q4. Explain why series resonance gives 'voltage magnification' and parallel resonance 'current magnification'.

At series resonance, current is maximum, producing reactive voltages across L and C equal to Q times input voltage ($V_L = V_C = QV$). At parallel resonance, circulating current between L and C equals Q times line current ($I_{\text{circ}} = QI$).

Q5. What is Brune's Test, and why is it conducted before interconnecting two-port networks?

Brune's test verifies whether interconnecting two-ports creates circulating loop currents that violate the port condition ($I_{\text{in}} = I_{\text{out}}$). If open-circuit ground potential is non-zero, an ideal $1:1$ isolation transformer is required.

Q6. How does winding resistance R_L affect practical parallel tank resonant frequency?

Winding resistance lowers resonant frequency below the ideal value: $\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}} = \omega_0\sqrt{1 - \frac{R_L^2 C}{L}}$. If $R_L \geq \sqrt{L/C}$, resonance cannot occur.

Q7. State the relationship between resonant frequency ω_0 and half-power frequencies in series resonance.

ω_0 is the exact geometric mean of half-power frequencies: $\omega_0 = \sqrt{\omega_1\omega_2}$. For high-$Q$ circuits ($Q \geq 10$), it is also approximately their arithmetic mean: $\omega_0 \approx (\omega_1 + \omega_2)/2$.

Q8. Why does a constant-k filter have undesirable impedance characteristics in its pass band?

Its characteristic impedance $Z_{0T}(f) = R_0\sqrt{1 - (f/f_c)^2}$ drops to zero at f_c, while $Z_{0\pi}(f)$ approaches infinity. This impedance mismatch causes large reflection losses near cut-off when terminated in R_0.

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Frequently Asked / Viva Questions (contd.)

Q9. What is the physical meaning of poles in a network transfer function?

Poles represent the complex natural frequencies (modes) of the system. The real part dictates exponential decay or growth rate, while the imaginary part specifies unforced transient oscillation frequency.

Q10. State the relationship between convolution in time and transfer function in frequency domain.

In the time domain, response is the convolution of input with the impulse response: $y(t) = x(t) * h(t)$. In the s-domain, this integral simplifies into algebraic multiplication: $Y(s) = H(s) \cdot X(s)$.

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Summary Matrix: Comprehensive Comparison of Two-Port Parameters and Resonant Circuits

Parameter / Circuit	Key Governing Relations	Reciprocity / Symmetry	Circuit Application
Z-Parameters $V_2 = z_{21}I_1 + z_{22}I_2$ $z_{11} = z_{22}$	$V_1 = z_{11}I_1 + z_{12}I_2$ $z_{12} = z_{21}$ Open-circuit test; series-series connections.		
Y-Parameters $I_2 = y_{21}V_1 + y_{22}V_2$ $y_{11} = y_{22}$	$I_1 = y_{11}V_1 + y_{12}V_2$ $y_{12} = y_{21}$ Short-circuit test; parallel-parallel connections.		
$ABCD$-Parameters $I_1 = CV_2 - DI_2$ $A = D$	$V_1 = AV_2 - BI_2$ $AD - BC = 1$ Transmission lines; cascaded stage networks.		
h-Parameters $I_2 = h_{21}I_1 + h_{22}V_2$ $\Delta h = 1$	$V_1 = h_{11}I_1 + h_{12}V_2$ $h_{12} = -h_{21}$ Transistor hybrid small-signal equivalent circuits.		
Series Resonant Circuit $Q = \omega_0 L/R$ (Resonance)	$\omega_0 = 1/\sqrt{LC},\ Z_{\text{min}} = R$ $V_L = V_C = QV$ Wave traps; radio tuners; voltage magnification.		
Parallel Resonant Tank $Z_d = \frac{L}{CR_L}$ (Max Z) (Anti-resonance)	$\omega_r = \sqrt{\frac{1}{LC} - \frac{R_L^2}{L^2}}$ $I_{\text{circ}} = QI_{\text{line}}$ RF tank circuits; oscillators; high-impedance loads.		

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